

VERIFICATION AND VALIDATION SPECIFICATION

Verification and Validation Check Matrix

Natural-Language Treatment Planning System (NL-TPS)

Source baseline: Controlled NL-TPS suite v0.1 including MQA-PKG-01

Document version: 0.1 - Proposed baseline

Date: 18 August 2026

Status: Discussion Draft - Planned checks, not executed results

Coverage: 2,144 unique checks: 1,952 requirement or interface checks and 192 component or subassembly checks

STATUS AND SAFETY NOTICE

This document specifies required checks. It does not claim that verification, validation, commissioning, clinical evaluation, acceptance, or release has occurred. A check becomes PASS only after controlled execution, attributable review, anomaly disposition, and approval by the required independent and clinical authorities.

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7.7	Plan review, selection, QA, and transfer (REV) - 27 checks	158
7.8	Data, interoperability, provenance, and audit (DAT) - 30 checks	165
7.9	AI and inference-model lifecycle (AIM) - 24 checks	173
7.10	Human factors, usability, and competency (HFE) - 18 checks	179
7.11	Privacy, cybersecurity, and secure development (SEC) - 24 checks	183
7.12	Clinical operations, monitoring, and resilience (OPS) - 21 checks	189
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8.4	Evidence and literature governance (EVD) - 72 checks	263
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8.6	Treatment-plan generation and calculation (PLN) - 108 checks	301
8.7	Plan review, selection, QA, and transfer (REV) - 81 checks	328

8.8 Data, interoperability, provenance, and audit (DAT) - 90 checks	349
8.9 AI and inference-model lifecycle (AIM) - 72 checks	371
8.10 Human factors, usability, and competency (HFE) - 54 checks	389
8.11 Privacy, cybersecurity, and secure development (SEC) - 72 checks	403
8.12 Clinical operations, monitoring, and resilience (OPS) - 63 checks	421
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Document control

Field	Value
Document owner	NL-TPS Program; verification authority assigned to T-VV; clinical validation authority remains with approved clinical governance, qualified medical physicists, radiation oncologists, dosimetrists, and other authorized specialists.
Source entities	2,088 unique entities in the controlled risk baseline plus 56 MQA-PKG-01 realization entities. Categorized F/NF/O requirement IDs remain aliases to canonical source IDs and do not create duplicate checks.
Check status	All rows are PLANNED and UNEXECUTED until an approved protocol, environment, expected result, actual result, anomaly record, reviewer, and disposition are linked.
Change authority	Clinical governance, QMP technical authority, quality management, systems engineering, security/privacy, interface control, operations, affected specialty teams, and independent V&V.
Supersession	None

Executive summary

This specification assigns one uniquely identified V&V check to every controlled high-level requirement, detailed sub-requirement, high-level interface requirement, sub-interface requirement, component responsibility, and MQA-PKG-01 realization entity. Each row defines the minimum direct verification action, its validation contribution, objective acceptance evidence, accountable owner, inherited method, risk gate, and initial status.

NO INFERRED PASS OR CLINICAL AUTHORITY

Trace completeness is not verification, a successful verification is not clinical validation, and validation is not authorization to treat. No model, retrieval system, dashboard, document generator, or automated service may mark a check PASS, accept an anomaly or residual risk, commission a route, approve a plan, declare machine readiness, or authorize clinical release.

1 Purpose, scope, and governing principles

Verification asks whether the specified requirement, interface, or component responsibility was implemented correctly against its controlled definition. Validation asks whether the integrated result supports the approved intended use safely and effectively for representative users, patients, workflows, sites, modalities, machines, and environments. Both are required where applicable; neither may be replaced by document review alone when executable behavior or human interaction is involved.

The matrix uses the suite's existing evidence hierarchy and applicable institutional interpretations of AAPM task-group and medical-physics practice guidance, ISO 14971 risk controls, IEC 62304 lifecycle controls, IEC 62366-1 usability engineering, DICOM PS3 conformance, IHE-RO profiles, approved commissioning evidence, and disease-site literature. Source editions, applicability, local

policy, tolerances, expected values, datasets, and acceptance thresholds shall be configuration-controlled in each executed protocol rather than inferred from this matrix.

2 Coverage and roll-up architecture

Entity set	Checks	Check class	Initial status
System high-level requirements	119	VVC-HLR	Planned
Detailed sub-requirements	357	VVC-SUB	Planned
High-level interface requirements	357	VVC-HLIR	Planned
Sub-interface requirements	1,071	VVC-SIR	Planned
Core component responsibilities	45	VVC-COMP	Planned
Interface component responsibilities	61	VVC-ICOMP	Planned
F/NF/O category component responsibilities	78	VVC-CCOMP	Planned
MQA realization requirements	12	VVC-MQA-R	Planned
MQA sub-requirements	36	VVC-MQA-S	Planned
MQA package subassemblies	8	VVC-MQA-C	Planned
TOTAL	2,144	One check per unique entity	Planned

A parent check does not pass by arithmetic alone. Every child must pass, but the parent must also demonstrate its emergent end-to-end outcome. An HLR therefore requires its three detailed children, three HLIRs, their SIRs and allocated components, plus direct system validation. An HLIR requires all three SIRs and every assigned interface-family profile. A component requires direct contract, unit, integration, fault, monitoring, recovery, allocation-coverage, and representative-workflow evidence.

3 Methods, independence, and execution record

Code	Method	Minimum evidence
I	Inspection	Approved requirement, design, code or configuration review; bidirectional trace; signature; checklist; version and integrity evidence.
T	Test	Controlled inputs, expected outputs, actual outputs, tolerances and sources, positive/negative/boundary/failure cases, logs, repeatability, and anomaly disposition.
A	Analysis	Reproducible calculation, coverage, uncertainty, statistical or hazard analysis with locked data, assumptions, software, reviewer, and conclusion.
D	Demonstration	Witnessed execution of the approved workflow in the controlled environment with attributable observations and outcome.

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Code	Method	Minimum evidence
HFE	Human-factors evaluation	Representative users, critical tasks, use-related hazards, environments, interruptions, workload, comprehension, recovery, observations, and acceptance decision.

Every executed check record shall include: immutable check and entity IDs; parent and allocation trace; approved protocol and revision; risk controls and hazard links; site, system, interface, machine, modality, technique, operating mode, software, model, evidence, and configuration versions; data manifest and hashes; preconditions; expected result and tolerance source; actual result; raw evidence and logs; deviations and anomalies; tester; independent reviewer; clinical/QMP validator where applicable; date and environment; pass/fail/blocked disposition; residual-risk decision; regression triggers; and release linkage.

4 Pass, fail, block, and change rules

- **PASS** requires every mandatory expected result to be met and every required artifact and approval to be present. A partial pass, average pass, model confidence, or undocumented expert judgment is not PASS.
- **FAIL** records any unmet expected result, unsafe or unexplained behavior, missing mandatory evidence, or unapproved variance. Failed tests are preserved; they are never overwritten by a retest.
- **BLOCKED** records a test that cannot validly execute because an approved prerequisite, environment, reference, authority, source, configuration, or dependency is unavailable or inconsistent. BLOCKED is not PASS.
- Every anomaly shall identify affected requirements, components, hazards, patients or records if any, containment, correction, retest, regression scope, root cause, CAPA when applicable, and authorized closure.
- Any change to intended use, requirement, risk control, source evidence, tolerance, interface, vendor/version, machine, model, data distribution, workflow, security control, dependency, monitoring threshold, or recovery behavior triggers documented impact analysis and risk-based regression.
- Sampling of entity IDs is prohibited. Statistical sampling within a parameter space is permitted only when the approved protocol defines the population, rationale, power or coverage basis, boundaries, rare and adverse cases, and residual uncertainty.

5 System high-level requirement V&V checks

Each HLR check verifies the parent outcome in addition to child closure and validates the integrated clinical or operational result.

5.1 Governance, intended use, and scope (GOV) - 7 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR GOV-001	Close direct-child coverage, then use controlled inspection, documented analysis to verify the parent outcome under the approved baseline. Challenge scope, authority, separation of duties, effective configuration, change, and release and every linked risk control.	Representative authorized roles execute an end-to-end governed scope, change, approval, and release scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; High; Planned
VVC-HLR GOV-002	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge scope, authority, separation of duties, effective configuration, change, and release and every linked risk control.	Representative authorized roles execute an end-to-end governed scope, change, approval, and release scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-HLR GOV-003	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge scope, authority, separation of duties, effective configuration, change, and release and every linked risk control.	Representative authorized roles execute an end-to-end governed scope, change, approval, and release scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/I; High; Planned
VVC-HLR GOV-004	Close direct-child coverage, then use objective testing, witnessed demonstration to verify the parent outcome under the approved baseline. Challenge scope, authority, separation of duties, effective configuration, change, and release and every linked risk control.	Representative authorized roles execute an end-to-end governed scope, change, approval, and release scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/D; High; Planned
VVC-HLR GOV-005	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge scope, authority, separation of duties, effective configuration, change, and release and every linked risk control.	Representative authorized roles execute an end-to-end governed scope, change, approval, and release scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/I; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR GOV-006	Close direct-child coverage, then use controlled inspection, documented analysis to verify the parent outcome under the approved baseline. Challenge scope, authority, separation of duties, effective configuration, change, and release and every linked risk control.	Representative authorized roles execute an end-to-end governed scope, change, approval, and release scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; High; Planned
VVC-HLR GOV-007	Close direct-child coverage, then use controlled inspection, documented analysis to verify the parent outcome under the approved baseline. Challenge scope, authority, separation of duties, effective configuration, change, and release and every linked risk control.	Representative authorized roles execute an end-to-end governed scope, change, approval, and release scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; High; Planned

5.2 Safety boundaries and human authority (SAF) - 10 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR SAF-001	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and every linked risk control.	Representative authorized roles execute an end-to-end safety-critical clinical workflow with wrong-context and prohibited-action challenges in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/I; Gate; Planned
VVC-HLR SAF-002	Close direct-child coverage, then use objective testing, witnessed demonstration to verify the parent outcome under the approved baseline. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and every linked risk control.	Representative authorized roles execute an end-to-end safety-critical clinical workflow with wrong-context and prohibited-action challenges in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SAF-003	Close direct-child coverage, then use objective testing, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and every linked risk control.	Representative authorized roles execute an end-to-end safety-critical clinical workflow with wrong-context and prohibited-action challenges in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/HFE; Gate; Planned
VVC-HLR				
SAF-004	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and every linked risk control.	Representative authorized roles execute an end-to-end safety-critical clinical workflow with wrong-context and prohibited-action challenges in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; Gate; Planned
VVC-HLR				
SAF-005	Close direct-child coverage, then use objective testing, witnessed demonstration to verify the parent outcome under the approved baseline. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and every linked risk control.	Representative authorized roles execute an end-to-end safety-critical clinical workflow with wrong-context and prohibited-action challenges in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; High; Planned
VVC-HLR				
SAF-006	Close direct-child coverage, then use objective testing, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and every linked risk control.	Representative authorized roles execute an end-to-end safety-critical clinical workflow with wrong-context and prohibited-action challenges in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/HFE; Gate; Planned
VVC-HLR				
SAF-007	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and every linked risk control.	Representative authorized roles execute an end-to-end safety-critical clinical workflow with wrong-context and prohibited-action challenges in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SAF-008	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and every linked risk control.	Representative authorized roles execute an end-to-end safety-critical clinical workflow with wrong-context and prohibited-action challenges in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Gate; Planned
VVC-HLR SAF-009	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and every linked risk control.	Representative authorized roles execute an end-to-end safety-critical clinical workflow with wrong-context and prohibited-action challenges in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Gate; Planned
VVC-HLR SAF-010	Close direct-child coverage, then use objective testing, witnessed demonstration to verify the parent outcome under the approved baseline. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and every linked risk control.	Representative authorized roles execute an end-to-end safety-critical clinical workflow with wrong-context and prohibited-action challenges in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; High; Planned

5.3 Natural-language and voice interaction (NLI) - 8 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR NLI-001	Close direct-child coverage, then use objective testing, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and every linked risk control.	Representative authorized roles execute an end-to-end typed and push-to-talk workflow containing realistic ambiguity and critical tokens in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/HFE; Gate; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
NLI-002	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and every linked risk control.	Representative authorized roles execute an end-to-end typed and push-to-talk workflow containing realistic ambiguity and critical tokens in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/I; Gate; Planned
VVC-HLR				
NLI-003	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and every linked risk control.	Representative authorized roles execute an end-to-end typed and push-to-talk workflow containing realistic ambiguity and critical tokens in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/I; High; Planned
VVC-HLR				
NLI-004	Close direct-child coverage, then use objective testing, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and every linked risk control.	Representative authorized roles execute an end-to-end typed and push-to-talk workflow containing realistic ambiguity and critical tokens in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/HFE; High; Planned
VVC-HLR				
NLI-005	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and every linked risk control.	Representative authorized roles execute an end-to-end typed and push-to-talk workflow containing realistic ambiguity and critical tokens in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; Gate; Planned
VVC-HLR				
NLI-006	Close direct-child coverage, then use objective testing, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and every linked risk control.	Representative authorized roles execute an end-to-end typed and push-to-talk workflow containing realistic ambiguity and critical tokens in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/HFE; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
NLI-007	Close direct-child coverage, then use objective testing, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and every linked risk control.	Representative authorized roles execute an end-to-end typed and push-to-talk workflow containing realistic ambiguity and critical tokens in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/HFE; Gate; Planned
VVC-HLR NLI-008	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and every linked risk control.	Representative authorized roles execute an end-to-end typed and push-to-talk workflow containing realistic ambiguity and critical tokens in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/I; High; Planned

5.4 Evidence and literature governance (EVD) - 8 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR EVD-001	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and every linked risk control.	Representative authorized roles execute an end-to-end clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-HLR EVD-002	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and every linked risk control.	Representative authorized roles execute an end-to-end clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
EVD-003	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and every linked risk control.	Representative authorized roles execute an end-to-end clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-HLR				
EVD-004	Close direct-child coverage, then use objective testing, witnessed demonstration to verify the parent outcome under the approved baseline. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and every linked risk control.	Representative authorized roles execute an end-to-end clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/D; Gate; Planned
VVC-HLR				
EVD-005	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and every linked risk control.	Representative authorized roles execute an end-to-end clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-HLR				
EVD-006	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and every linked risk control.	Representative authorized roles execute an end-to-end clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; Gate; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
EVD-007	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and non-applicable-rule behavior and every linked risk control.	Representative authorized roles execute an end-to-end clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-HLR				
EVD-008	Close direct-child coverage, then use objective testing, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and non-applicable-rule behavior and every linked risk control.	Representative authorized roles execute an end-to-end clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/HFE; High; Planned

5.5 Clinical context, imaging, registration, and contours (CLN) - 9 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR				
CLN-001	Close direct-child coverage, then use objective testing, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and every linked risk control.	Representative authorized roles execute an end-to-end representative imaging, registration, contouring, and anatomy-review case in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/HFE; High; Planned
VVC-HLR				
CLN-002	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and every linked risk control.	Representative authorized roles execute an end-to-end representative imaging, registration, contouring, and anatomy-review case in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/I; Gate; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
CLN-003	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and every linked risk control.	Representative authorized roles execute an end-to-end representative imaging, registration, contouring, and anatomy-review case in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; Gate; Planned
VVC-HLR				
CLN-004	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and every linked risk control.	Representative authorized roles execute an end-to-end representative imaging, registration, contouring, and anatomy-review case in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/I; High; Planned
VVC-HLR				
CLN-005	Close direct-child coverage, then use objective testing, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and every linked risk control.	Representative authorized roles execute an end-to-end representative imaging, registration, contouring, and anatomy-review case in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/HFE; High; Planned
VVC-HLR				
CLN-006	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and every linked risk control.	Representative authorized roles execute an end-to-end representative imaging, registration, contouring, and anatomy-review case in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; High; Planned
VVC-HLR				
CLN-007	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and every linked risk control.	Representative authorized roles execute an end-to-end representative imaging, registration, contouring, and anatomy-review case in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/I; Gate; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
CLN-008	Close direct-child coverage, then use witnessed demonstration, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and every linked risk control.	Representative authorized roles execute an end-to-end representative imaging, registration, contouring, and anatomy-review case in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; D/HFE; High; Planned
VVC-HLR CLN-009	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and every linked risk control.	Representative authorized roles execute an end-to-end representative imaging, registration, contouring, and anatomy-review case in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/I; High; Planned

5.6 Treatment-plan generation and calculation (PLN) - 12 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR PLN-001	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and every linked risk control.	Representative authorized roles execute an end-to-end representative treatment-planning case spanning intent, candidates, dose, and robustness in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/I; Gate; Planned
VVC-HLR PLN-002	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and every linked risk control.	Representative authorized roles execute an end-to-end representative treatment-planning case spanning intent, candidates, dose, and robustness in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
PLN-003	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and every linked risk control.	Representative authorized roles execute an end-to-end representative treatment-planning case spanning intent, candidates, dose, and robustness in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-HLR				
PLN-004	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and every linked risk control.	Representative authorized roles execute an end-to-end representative treatment-planning case spanning intent, candidates, dose, and robustness in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/I; Gate; Planned
VVC-HLR				
PLN-005	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and every linked risk control.	Representative authorized roles execute an end-to-end representative treatment-planning case spanning intent, candidates, dose, and robustness in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; High; Planned
VVC-HLR				
PLN-006	Close direct-child coverage, then use objective testing, witnessed demonstration to verify the parent outcome under the approved baseline. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and every linked risk control.	Representative authorized roles execute an end-to-end representative treatment-planning case spanning intent, candidates, dose, and robustness in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; High; Planned
VVC-HLR				
PLN-007	Close direct-child coverage, then use controlled inspection, documented analysis to verify the parent outcome under the approved baseline. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and every linked risk control.	Representative authorized roles execute an end-to-end representative treatment-planning case spanning intent, candidates, dose, and robustness in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
PLN-008	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and every linked risk control.	Representative authorized roles execute an end-to-end representative treatment-planning case spanning intent, candidates, dose, and robustness in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; High; Planned
VVC-HLR PLN-009	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and every linked risk control.	Representative authorized roles execute an end-to-end representative treatment-planning case spanning intent, candidates, dose, and robustness in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/I; Gate; Planned
VVC-HLR PLN-010	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and every linked risk control.	Representative authorized roles execute an end-to-end representative treatment-planning case spanning intent, candidates, dose, and robustness in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; Gate; Planned
VVC-HLR PLN-011	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and every linked risk control.	Representative authorized roles execute an end-to-end representative treatment-planning case spanning intent, candidates, dose, and robustness in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; Gate; Planned
VVC-HLR PLN-012	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and every linked risk control.	Representative authorized roles execute an end-to-end representative treatment-planning case spanning intent, candidates, dose, and robustness in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned

5.7 Plan review, selection, QA, and transfer (REV) - 9 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR REV-001	Close direct-child coverage, then use witnessed demonstration, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; D/HFE; High; Planned
VVC-HLR REV-002	Close direct-child coverage, then use witnessed demonstration, objective testing to verify the parent outcome under the approved baseline. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; D/T; Gate; Planned
VVC-HLR REV-003	Close direct-child coverage, then use controlled inspection, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/HFE; High; Planned
VVC-HLR REV-004	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
REV-005	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/I; High; Planned
VVC-HLR				
REV-006	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/I; Gate; Planned
VVC-HLR				
REV-007	Close direct-child coverage, then use objective testing, witnessed demonstration to verify the parent outcome under the approved baseline. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/D; High; Planned
VVC-HLR				
REV-008	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; High; Planned
VVC-HLR				
REV-009	Close direct-child coverage, then use controlled inspection, documented analysis to verify the parent outcome under the approved baseline. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; High; Planned

5.8 Data, interoperability, provenance, and audit (DAT) - 10 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR DAT-001	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end clinical-object lifecycle and round-trip data exchange workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-HLR DAT-002	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end clinical-object lifecycle and round-trip data exchange workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-HLR DAT-003	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end clinical-object lifecycle and round-trip data exchange workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; Gate; Planned
VVC-HLR DAT-004	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end clinical-object lifecycle and round-trip data exchange workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
DAT-005	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end clinical-object lifecycle and round-trip data exchange workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-HLR				
DAT-006	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end clinical-object lifecycle and round-trip data exchange workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-HLR				
DAT-007	Close direct-child coverage, then use objective testing to verify the parent outcome under the approved baseline. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end clinical-object lifecycle and round-trip data exchange workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T; High; Planned
VVC-HLR				
DAT-008	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end clinical-object lifecycle and round-trip data exchange workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-HLR				
DAT-009	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end clinical-object lifecycle and round-trip data exchange workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/I; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
DAT-010	Close direct-child coverage, then use controlled inspection, documented analysis to verify the parent outcome under the approved baseline. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end clinical-object lifecycle and round-trip data exchange workflow in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned

5.9 AI and inference-model lifecycle (AIM) - 8 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR AIM-001	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and every linked risk control.	Representative authorized roles execute an end-to-end model-supported clinical draft workflow including abstention, failure, and monitoring cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T; High; Planned
VVC-HLR AIM-002	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and every linked risk control.	Representative authorized roles execute an end-to-end model-supported clinical draft workflow including abstention, failure, and monitoring cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/A; High; Planned
VVC-HLR AIM-003	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and every linked risk control.	Representative authorized roles execute an end-to-end model-supported clinical draft workflow including abstention, failure, and monitoring cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/A; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
AIM-004	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and every linked risk control.	Representative authorized roles execute an end-to-end model-supported clinical draft workflow including abstention, failure, and monitoring cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T; High; Planned
VVC-HLR				
AIM-005	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and every linked risk control.	Representative authorized roles execute an end-to-end model-supported clinical draft workflow including abstention, failure, and monitoring cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T; High; Planned
VVC-HLR				
AIM-006	Close direct-child coverage, then use controlled inspection, documented analysis to verify the parent outcome under the approved baseline. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and every linked risk control.	Representative authorized roles execute an end-to-end model-supported clinical draft workflow including abstention, failure, and monitoring cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-HLR				
AIM-007	Close direct-child coverage, then use documented analysis, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and every linked risk control.	Representative authorized roles execute an end-to-end model-supported clinical draft workflow including abstention, failure, and monitoring cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; A/HFE; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
AIM-008	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and every linked risk control.	Representative authorized roles execute an end-to-end model-supported clinical draft workflow including abstention, failure, and monitoring cases in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T; High; Planned

5.10 Human factors, usability, and competency (HFE) - 6 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR HFE-001	Close direct-child coverage, then use representative-user human-factors evaluation, objective testing to verify the parent outcome under the approved baseline. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and every linked risk control.	Representative authorized roles execute an end-to-end representative critical-task usability scenario with interruptions and recovery in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; HFE/T; Gate; Planned
VVC-HLR HFE-002	Close direct-child coverage, then use representative-user human-factors evaluation, controlled inspection to verify the parent outcome under the approved baseline. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and every linked risk control.	Representative authorized roles execute an end-to-end representative critical-task usability scenario with interruptions and recovery in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; HFE/I; High; Planned
VVC-HLR HFE-003	Close direct-child coverage, then use representative-user human-factors evaluation, controlled inspection to verify the parent outcome under the approved baseline. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and every linked risk control.	Representative authorized roles execute an end-to-end representative critical-task usability scenario with interruptions and recovery in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; HFE/I; Moderate; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
HFE-004	Close direct-child coverage, then use representative-user human-factors evaluation, documented analysis to verify the parent outcome under the approved baseline. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and every linked risk control.	Representative authorized roles execute an end-to-end representative critical-task usability scenario with interruptions and recovery in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; HFE/A; High; Planned
VVC-HLR HFE-005	Close direct-child coverage, then use representative-user human-factors evaluation, controlled inspection to verify the parent outcome under the approved baseline. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and every linked risk control.	Representative authorized roles execute an end-to-end representative critical-task usability scenario with interruptions and recovery in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; HFE/I; High; Planned
VVC-HLR HFE-006	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and every linked risk control.	Representative authorized roles execute an end-to-end representative critical-task usability scenario with interruptions and recovery in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; High; Planned

5.11 Privacy, cybersecurity, and secure development (SEC) - 8 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR SEC-001	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and every linked risk control.	Representative authorized roles execute an end-to-end threat, privacy, access-control, and incident-response scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SEC-002	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and every linked risk control.	Representative authorized roles execute an end-to-end threat, privacy, access-control, and incident-response scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ T; High; Planned
VVC-HLR				
SEC-003	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and every linked risk control.	Representative authorized roles execute an end-to-end threat, privacy, access-control, and incident-response scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/ I; High; Planned
VVC-HLR				
SEC-004	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and every linked risk control.	Representative authorized roles execute an end-to-end threat, privacy, access-control, and incident-response scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/ A; High; Planned
VVC-HLR				
SEC-005	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and every linked risk control.	Representative authorized roles execute an end-to-end threat, privacy, access-control, and incident-response scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/ I; High; Planned
VVC-HLR				
SEC-006	Close direct-child coverage, then use controlled inspection, documented analysis to verify the parent outcome under the approved baseline. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and every linked risk control.	Representative authorized roles execute an end-to-end threat, privacy, access-control, and incident-response scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ A; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SEC-007	Close direct-child coverage, then use objective testing, controlled inspection to verify the parent outcome under the approved baseline. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and every linked risk control.	Representative authorized roles execute an end-to-end threat, privacy, access-control, and incident-response scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/I; High; Planned
VVC-HLR SEC-008	Close direct-child coverage, then use witnessed demonstration, controlled inspection to verify the parent outcome under the approved baseline. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and every linked risk control.	Representative authorized roles execute an end-to-end threat, privacy, access-control, and incident-response scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; D/I; High; Planned

5.12 Clinical operations, monitoring, and resilience (OPS) - 7 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR OPS-001	Close direct-child coverage, then use objective testing, documented analysis to verify the parent outcome under the approved baseline. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end production-like load, dependency failure, degraded-mode, and recovery scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; High; Planned
VVC-HLR OPS-002	Close direct-child coverage, then use objective testing to verify the parent outcome under the approved baseline. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end production-like load, dependency failure, degraded-mode, and recovery scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
OPS-003	Close direct-child coverage, then use witnessed demonstration, objective testing to verify the parent outcome under the approved baseline. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end production-like load, dependency failure, degraded-mode, and recovery scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; D/T; High; Planned
VVC-HLR OPS-004	Close direct-child coverage, then use witnessed demonstration, objective testing to verify the parent outcome under the approved baseline. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end production-like load, dependency failure, degraded-mode, and recovery scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; D/T; High; Planned
VVC-HLR OPS-005	Close direct-child coverage, then use witnessed demonstration, controlled inspection to verify the parent outcome under the approved baseline. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end production-like load, dependency failure, degraded-mode, and recovery scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; D/I; High; Planned
VVC-HLR OPS-006	Close direct-child coverage, then use controlled inspection, witnessed demonstration to verify the parent outcome under the approved baseline. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end production-like load, dependency failure, degraded-mode, and recovery scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/D; High; Planned
VVC-HLR OPS-007	Close direct-child coverage, then use controlled inspection, documented analysis to verify the parent outcome under the approved baseline. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and every linked risk control.	Representative authorized roles execute an end-to-end production-like load, dependency failure, degraded-mode, and recovery scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; High; Planned

5.13 Verification, commissioning, release, and change control (VAL) - 9 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR VAL-001	Close direct-child coverage, then use controlled inspection, documented analysis to verify the parent outcome under the approved baseline. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and every linked risk control.	Representative authorized roles execute an end-to-end controlled reference-case, anomaly, regression, and release-decision scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-HLR VAL-002	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and every linked risk control.	Representative authorized roles execute an end-to-end controlled reference-case, anomaly, regression, and release-decision scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-HLR VAL-003	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and every linked risk control.	Representative authorized roles execute an end-to-end controlled reference-case, anomaly, regression, and release-decision scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; Gate; Planned
VVC-HLR VAL-004	Close direct-child coverage, then use objective testing, witnessed demonstration to verify the parent outcome under the approved baseline. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and every linked risk control.	Representative authorized roles execute an end-to-end controlled reference-case, anomaly, regression, and release-decision scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/ D; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VAL-005	Close direct-child coverage, then use documented analysis, representative-user human-factors evaluation to verify the parent outcome under the approved baseline. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and every linked risk control.	Representative authorized roles execute an end-to-end controlled reference-case, anomaly, regression, and release-decision scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; A/HFE; Gate; Planned
VVC-HLR				
VAL-006	Close direct-child coverage, then use controlled inspection, documented analysis to verify the parent outcome under the approved baseline. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and every linked risk control.	Representative authorized roles execute an end-to-end controlled reference-case, anomaly, regression, and release-decision scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-HLR				
VAL-007	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and every linked risk control.	Representative authorized roles execute an end-to-end controlled reference-case, anomaly, regression, and release-decision scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-HLR				
VAL-008	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and every linked risk control.	Representative authorized roles execute an end-to-end controlled reference-case, anomaly, regression, and release-decision scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-HLR				
VAL-009	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and every linked risk control.	Representative authorized roles execute an end-to-end controlled reference-case, anomaly, regression, and release-decision scenario in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned

5.14 Accreditation-profile and clinical-quality governance (ACC) - 8 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLR ACC-001	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and every linked risk control.	Representative authorized roles execute an end-to-end QMP-led quality and accreditation-evidence workflow without system-generated compliance claims in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-HLR ACC-002	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and every linked risk control.	Representative authorized roles execute an end-to-end QMP-led quality and accreditation-evidence workflow without system-generated compliance claims in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-HLR ACC-003	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and every linked risk control.	Representative authorized roles execute an end-to-end QMP-led quality and accreditation-evidence workflow without system-generated compliance claims in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-HLR ACC-004	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and every linked risk control.	Representative authorized roles execute an end-to-end QMP-led quality and accreditation-evidence workflow without system-generated compliance claims in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-HLR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
ACC-005	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and every linked risk control.	Representative authorized roles execute an end-to-end QMP-led quality and accreditation-evidence workflow without system-generated compliance claims in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-HLR				
ACC-006	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and every linked risk control.	Representative authorized roles execute an end-to-end QMP-led quality and accreditation-evidence workflow without system-generated compliance claims in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-HLR				
ACC-007	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and every linked risk control.	Representative authorized roles execute an end-to-end QMP-led quality and accreditation-evidence workflow without system-generated compliance claims in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-HLR				
ACC-008	Close direct-child coverage, then use controlled inspection, objective testing to verify the parent outcome under the approved baseline. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and every linked risk control.	Representative authorized roles execute an end-to-end QMP-led quality and accreditation-evidence workflow without system-generated compliance claims in the intended-use environment and confirm clinical suitability, human authority, comprehensibility, and recovery.	All three detailed-child and three HLIR checks pass, and the direct parent outcome is observed end to end. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned

6 Detailed sub-requirement V&V checks

Each detailed child receives a direct executable or inspectable check through its allocated components; source and F/NF/O alias IDs share one result.

6.1 Governance, intended use, and scope (GOV) - 21 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB GOV-001-01	Use controlled inspection, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/HFE; High; Planned
VVC-SUB GOV-001-02	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/I; High; Planned
VVC-SUB GOV-001-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; High; Planned
VVC-SUB GOV-002-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
GOV-002-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T; High; Planned
VVC-SUB				
GOV-002-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SUB				
GOV-003-01	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T; Moderate; Planned
VVC-SUB				
GOV-003-02	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ A; High; Planned
VVC-SUB				
GOV-003-03	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/I; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
GOV-004-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SUB				
GOV-004-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/HFE; High; Planned
VVC-SUB				
GOV-004-03	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/I; High; Planned
VVC-SUB				
GOV-005-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SUB				
GOV-005-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB GOV-005-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SUB GOV-006-01	Use controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I; High; Planned
VVC-SUB GOV-006-02	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/I; High; Planned
VVC-SUB GOV-006-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; High; Planned
VVC-SUB GOV-007-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB GOV-007-02	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/I; High; Planned
VVC-SUB GOV-007-03	Use controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to scope, authority, separation of duties, effective configuration, change, and release.	Demonstrate this child within a representative governed scope, change, approval, and release scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I; High; Planned

6.2 Safety boundaries and human authority (SAF) - 30 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB SAF-001-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; Gate; Planned
VVC-SUB SAF-001-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SAF-001-03	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; High; Planned
VVC-SUB				
SAF-002-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Gate; Planned
VVC-SUB				
SAF-002-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/HFE; High; Planned
VVC-SUB				
SAF-002-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Moderate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SAF-003-01	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/HFE; Gate; Planned
VVC-SUB				
SAF-003-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T; Gate; Planned
VVC-SUB				
SAF-003-03	Use controlled inspection, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/HFE; Gate; Planned
VVC-SUB				
SAF-004-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SAF-004-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/HFE; High; Planned
VVC-SUB				
SAF-004-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; High; Planned
VVC-SUB				
SAF-005-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; High; Planned
VVC-SUB				
SAF-005-02	Use witnessed demonstration, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; D/HFE; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SAF-005-03	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T; High; Planned
VVC-SUB				
SAF-006-01	Use controlled inspection, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/HFE; High; Planned
VVC-SUB				
SAF-006-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/HFE; High; Planned
VVC-SUB				
SAF-006-03	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SAF-007-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; High; Planned
VVC-SUB				
SAF-007-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/ A; Gate; Planned
VVC-SUB				
SAF-007-03	Use witnessed demonstration, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; D/ I; High; Planned
VVC-SUB				
SAF-008-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ A; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SAF-008-02	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; Gate; Planned
VVC-SUB				
SAF-008-03	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/ A; High; Planned
VVC-SUB				
SAF-009-01	Use controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I; Gate; Planned
VVC-SUB				
SAF-009-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SAF-009-03	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/I; High; Planned
VVC-SUB				
SAF-010-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-SUB				
SAF-010-02	Use witnessed demonstration, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; D/T; High; Planned
VVC-SUB				
SAF-010-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Demonstrate this child within a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Moderate; Planned

6.3 Natural-language and voice interaction (NLI) - 24 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB NLI-001-01	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/HFE; High; Planned
VVC-SUB NLI-001-02	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/ T; High; Planned
VVC-SUB NLI-001-03	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/HFE; Gate; Planned
VVC-SUB NLI-002-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/ T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
NLI-002-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T; Gate; Planned
VVC-SUB				
NLI-002-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I; T; High; Planned
VVC-SUB				
NLI-003-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I; T; High; Planned
VVC-SUB				
NLI-003-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T; A; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
NLI-003-03	Use witnessed demonstration, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; D/I; High; Planned
VVC-SUB				
NLI-004-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SUB				
NLI-004-02	Use representative-user human-factors evaluation, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; HFE/I; High; Planned
VVC-SUB				
NLI-004-03	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
NLI-005-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; Gate; Planned
VVC-SUB				
NLI-005-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/HFE; High; Planned
VVC-SUB				
NLI-005-03	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SUB				
NLI-006-01	Use controlled inspection, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/HFE; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
NLI-006-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T; High; Planned
VVC-SUB				
NLI-006-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/ T; High; Planned
VVC-SUB				
NLI-007-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/ T; Gate; Planned
VVC-SUB				
NLI-007-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/HFE; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
NLI-007-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SUB				
NLI-008-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SUB				
NLI-008-02	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/I; High; Planned
VVC-SUB				
NLI-008-03	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation.	Demonstrate this child within a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T; High; Planned

6.4 Evidence and literature governance (EVD) - 24 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
EVD-001-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ T; High; Planned
VVC-SUB				
EVD-001-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T; High; Planned
VVC-SUB				
EVD-001-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ T; High; Planned
VVC-SUB				
EVD-002-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
EVD-002-02	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/I; High; Planned
VVC-SUB				
EVD-002-03	Use witnessed demonstration, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; D/HFE; High; Planned
VVC-SUB				
EVD-003-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SUB				
EVD-003-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
EVD-003-03	Use controlled inspection, witnessed demonstration to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/D; High; Planned
VVC-SUB				
EVD-004-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; Gate; Planned
VVC-SUB				
EVD-004-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T; Gate; Planned
VVC-SUB				
EVD-004-03	Use witnessed demonstration, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; D/I; Moderate; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
EVD-005-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ T; High; Planned
VVC-SUB				
EVD-005-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/ A; High; Planned
VVC-SUB				
EVD-005-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ T; Moderate; Planned
VVC-SUB				
EVD-006-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ T; Gate; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
EVD-006-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; Moderate; Planned
VVC-SUB				
EVD-006-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SUB				
EVD-007-01	Use controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I; High; Planned
VVC-SUB				
EVD-007-02	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/I; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
EVD-007-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SUB				
EVD-008-01	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T; High; Planned
VVC-SUB				
EVD-008-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/HFE; High; Planned
VVC-SUB				
EVD-008-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Demonstrate this child within a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned

6.5 Clinical context, imaging, registration, and contours (CLN) - 27 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB CLN-001-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; High; Planned
VVC-SUB CLN-001-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/HFE; Moderate; Planned
VVC-SUB CLN-001-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Moderate; Planned
VVC-SUB CLN-002-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
CLN-002-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; High; Planned
VVC-SUB				
CLN-002-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; High; Planned
VVC-SUB				
CLN-003-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-SUB				
CLN-003-02	Use witnessed demonstration, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; D/HFE; Gate; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
CLN-003-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/ T; Gate; Planned
VVC-SUB				
CLN-004-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/ T; High; Planned
VVC-SUB				
CLN-004-02	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/ I; High; Planned
VVC-SUB				
CLN-004-03	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
CLN-005-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/ T; High; Planned
VVC-SUB				
CLN-005-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/HFE; High; Planned
VVC-SUB				
CLN-005-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/ T; High; Planned
VVC-SUB				
CLN-006-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/ T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
CLN-006-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; High; Planned
VVC-SUB				
CLN-006-03	Use witnessed demonstration, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; D/I; High; Planned
VVC-SUB				
CLN-007-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-SUB				
CLN-007-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/HFE; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
CLN-007-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; High; Planned
VVC-SUB				
CLN-008-01	Use witnessed demonstration, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; D/HFE; High; Planned
VVC-SUB				
CLN-008-02	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/I; High; Planned
VVC-SUB				
CLN-008-03	Use representative-user human-factors evaluation, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; HFE/A; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
CLN-009-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; High; Planned
VVC-SUB CLN-009-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; High; Planned
VVC-SUB CLN-009-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Demonstrate this child within a representative representative imaging, registration, contouring, and anatomy-review case and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; High; Planned

6.6 Treatment-plan generation and calculation (PLN) - 36 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB PLN-001-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
PLN-001-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; Gate; Planned
VVC-SUB				
PLN-001-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Moderate; Planned
VVC-SUB				
PLN-002-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-SUB				
PLN-002-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T; Gate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
PLN-002-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SUB				
PLN-003-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; Gate; Planned
VVC-SUB				
PLN-003-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T; Gate; Planned
VVC-SUB				
PLN-003-03	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; Gate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
PLN-004-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-SUB				
PLN-004-02	Use witnessed demonstration, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; D/HFE; Moderate; Planned
VVC-SUB				
PLN-004-03	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; Gate; Planned
VVC-SUB				
PLN-005-01	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/I; Moderate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
PLN-005-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/HFE; Gate; Planned
VVC-SUB				
PLN-005-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SUB				
PLN-006-01	Use objective testing, witnessed demonstration to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-SUB				
PLN-006-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/HFE; Gate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
PLN-006-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Moderate; Planned
VVC-SUB				
PLN-007-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-SUB				
PLN-007-02	Use documented analysis, witnessed demonstration to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; A/D; High; Planned
VVC-SUB				
PLN-007-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; Moderate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
PLN-008-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Moderate; Planned
VVC-SUB				
PLN-008-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; Gate; Planned
VVC-SUB				
PLN-008-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SUB				
PLN-009-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
PLN-009-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T; High; Planned
VVC-SUB				
PLN-009-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-SUB				
PLN-010-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-SUB				
PLN-010-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; Gate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
PLN-010-03	Use witnessed demonstration, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; D/I; Gate; Planned
VVC-SUB				
PLN-011-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-SUB				
PLN-011-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; High; Planned
VVC-SUB				
PLN-011-03	Use controlled inspection, witnessed demonstration to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/D; Moderate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
PLN-012-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Moderate; Planned
VVC-SUB				
PLN-012-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T; Moderate; Planned
VVC-SUB				
PLN-012-03	Use controlled inspection, witnessed demonstration to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Demonstrate this child within a representative representative treatment-planning case spanning intent, candidates, dose, and robustness and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/D; Moderate; Planned

6.7 Plan review, selection, QA, and transfer (REV) - 27 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB				
REV-001-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; Moderate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
REV-001-02	Use witnessed demonstration, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; D/HFE; High; Planned
VVC-SUB				
REV-001-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; Moderate; Planned
VVC-SUB				
REV-002-01	Use controlled inspection, witnessed demonstration to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/D; Gate; Planned
VVC-SUB				
REV-002-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
REV-002-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; Moderate; Planned
VVC-SUB				
REV-003-01	Use controlled inspection, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/HFE; Moderate; Planned
VVC-SUB				
REV-003-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; High; Planned
VVC-SUB				
REV-003-03	Use witnessed demonstration, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; D/I; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
REV-004-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SUB				
REV-004-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/HFE; High; Planned
VVC-SUB				
REV-004-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; Moderate; Planned
VVC-SUB				
REV-005-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
REV-005-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T; High; Planned
VVC-SUB				
REV-005-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SUB				
REV-006-01	Use controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I; Gate; Planned
VVC-SUB				
REV-006-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
REV-006-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; Moderate; Planned
VVC-SUB				
REV-007-01	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T; High; Planned
VVC-SUB				
REV-007-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; High; Planned
VVC-SUB				
REV-007-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
REV-008-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SUB				
REV-008-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/ A; High; Planned
VVC-SUB				
REV-008-03	Use witnessed demonstration, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; D/I; Moderate; Planned
VVC-SUB				
REV-009-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ A; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
REV-009-02	Use documented analysis, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; A/T; High; Planned
VVC-SUB				
REV-009-03	Use controlled inspection, witnessed demonstration to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Demonstrate this child within a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/D; High; Planned

6.8 Data, interoperability, provenance, and audit (DAT) - 30 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB				
DAT-001-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SUB				
DAT-001-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T; Moderate; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB DAT-001-03	Use witnessed demonstration, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; D/I; Moderate; Planned
VVC-SUB DAT-002-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SUB DAT-002-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; Moderate; Planned
VVC-SUB DAT-002-03	Use witnessed demonstration, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; D/T; Moderate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
DAT-003-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-SUB				
DAT-003-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; Moderate; Planned
VVC-SUB				
DAT-003-03	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; High; Planned
VVC-SUB				
DAT-004-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
DAT-004-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T; Moderate; Planned
VVC-SUB				
DAT-004-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SUB				
DAT-005-01	Use controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I; High; Planned
VVC-SUB				
DAT-005-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
DAT-005-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-SUB				
DAT-006-01	Use controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I; High; Planned
VVC-SUB				
DAT-006-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; High; Planned
VVC-SUB				
DAT-006-03	Use controlled inspection, witnessed demonstration to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/D; Moderate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
DAT-007-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SUB				
DAT-007-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T; High; Planned
VVC-SUB				
DAT-007-03	Use witnessed demonstration, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; D/I; High; Planned
VVC-SUB				
DAT-008-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Moderate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
DAT-008-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T; High; Planned
VVC-SUB				
DAT-008-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SUB				
DAT-009-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SUB				
DAT-009-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
DAT-009-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SUB				
DAT-010-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; Moderate; Planned
VVC-SUB				
DAT-010-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T; Moderate; Planned
VVC-SUB				
DAT-010-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Demonstrate this child within a representative clinical-object lifecycle and round-trip data exchange workflow and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned

6.9 AI and inference-model lifecycle (AIM) - 24 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
AIM-001-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-SUB				
AIM-001-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T; High; Planned
VVC-SUB				
AIM-001-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-SUB				
AIM-002-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
AIM-002-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/A; High; Planned
VVC-SUB				
AIM-002-03	Use witnessed demonstration, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; D/I; Moderate; Planned
VVC-SUB				
AIM-003-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T; High; Planned
VVC-SUB				
AIM-003-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
AIM-003-03	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/A; High; Planned
VVC-SUB				
AIM-004-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T; High; Planned
VVC-SUB				
AIM-004-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T; High; Planned
VVC-SUB				
AIM-004-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
AIM-005-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-SUB				
AIM-005-02	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/ I; High; Planned
VVC-SUB				
AIM-005-03	Use controlled inspection, witnessed demonstration to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ D; High; Planned
VVC-SUB				
AIM-006-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ A; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
AIM-006-02	Use documented analysis, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; A/T; Moderate; Planned
VVC-SUB				
AIM-006-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SUB				
AIM-007-01	Use documented analysis, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; A/T; High; Planned
VVC-SUB				
AIM-007-02	Use witnessed demonstration, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; D/HFE; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
AIM-007-03	Use documented analysis, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; A/ I; High; Planned
VVC-SUB				
AIM-008-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ A; High; Planned
VVC-SUB				
AIM-008-02	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-SUB				
AIM-008-03	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Demonstrate this child within a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/ I; High; Planned

6.10 Human factors, usability, and competency (HFE) - 18 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB HFE-001-01	Use representative-user human-factors evaluation, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; HFE/T; Gate; Planned
VVC-SUB HFE-001-02	Use objective testing, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/HFE; Moderate; Planned
VVC-SUB HFE-001-03	Use representative-user human-factors evaluation, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; HFE/I; Moderate; Planned
VVC-SUB HFE-002-01	Use controlled inspection, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/HFE; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
HFE-002-02	Use representative-user human-factors evaluation, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; HFE/I; Moderate; Planned
VVC-SUB				
HFE-002-03	Use representative-user human-factors evaluation, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; HFE/A; Moderate; Planned
VVC-SUB				
HFE-003-01	Use controlled inspection, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/HFE; Moderate; Planned
VVC-SUB				
HFE-003-02	Use representative-user human-factors evaluation, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; HFE/T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
HFE-003-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SUB				
HFE-004-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; High; Planned
VVC-SUB				
HFE-004-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/A; Gate; Planned
VVC-SUB				
HFE-004-03	Use documented analysis, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; A/I; Moderate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
HFE-005-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SUB				
HFE-005-02	Use representative-user human-factors evaluation, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; HFE/I; High; Planned
VVC-SUB				
HFE-005-03	Use representative-user human-factors evaluation, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; HFE/A; High; Planned
VVC-SUB				
HFE-006-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
HFE-006-02	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/I; Moderate; Planned
VVC-SUB				
HFE-006-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Demonstrate this child within a representative representative critical-task usability scenario with interruptions and recovery and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; High; Planned

6.11 Privacy, cybersecurity, and secure development (SEC) - 24 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB				
SEC-001-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SUB				
SEC-001-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SEC-001-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SUB				
SEC-002-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SUB				
SEC-002-02	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/I; High; Planned
VVC-SUB				
SEC-002-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SEC-003-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ T; High; Planned
VVC-SUB				
SEC-003-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T; High; Planned
VVC-SUB				
SEC-003-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ T; Moderate; Planned
VVC-SUB				
SEC-004-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ A; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SEC-004-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T; High; Planned
VVC-SUB				
SEC-004-03	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SUB				
SEC-005-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SUB				
SEC-005-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SEC-005-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ T; High; Planned
VVC-SUB				
SEC-006-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ A; High; Planned
VVC-SUB				
SEC-006-02	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ T; High; Planned
VVC-SUB				
SEC-006-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ A; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SEC-007-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ T; High; Planned
VVC-SUB				
SEC-007-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T; High; Planned
VVC-SUB				
SEC-007-03	Use controlled inspection, witnessed demonstration to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ D; High; Planned
VVC-SUB				
SEC-008-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ A; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SEC-008-02	Use witnessed demonstration, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; D/T; High; Planned
VVC-SUB				
SEC-008-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Demonstrate this child within a representative threat, privacy, access-control, and incident-response scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned

6.12 Clinical operations, monitoring, and resilience (OPS) - 21 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB				
OPS-001-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; High; Planned
VVC-SUB				
OPS-001-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB OPS-001-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/ T; High; Planned
VVC-SUB OPS-002-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/ T; Moderate; Planned
VVC-SUB OPS-002-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T; Moderate; Planned
VVC-SUB OPS-002-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/ T; Moderate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
OPS-003-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-SUB				
OPS-003-02	Use witnessed demonstration, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; D/T; High; Planned
VVC-SUB				
OPS-003-03	Use witnessed demonstration, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; D/I; High; Planned
VVC-SUB				
OPS-004-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
OPS-004-02	Use witnessed demonstration, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; D/ T; High; Planned
VVC-SUB				
OPS-004-03	Use controlled inspection, witnessed demonstration to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/ D; Moderate; Planned
VVC-SUB				
OPS-005-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/ T; High; Planned
VVC-SUB				
OPS-005-02	Use witnessed demonstration, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; D/ T; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
OPS-005-03	Use controlled inspection, witnessed demonstration to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/D; High; Planned
VVC-SUB				
OPS-006-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SUB				
OPS-006-02	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; High; Planned
VVC-SUB				
OPS-006-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
OPS-007-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; High; Planned
VVC-SUB				
OPS-007-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; High; Planned
VVC-SUB				
OPS-007-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Demonstrate this child within a representative production-like load, dependency failure, degraded-mode, and recovery scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; High; Planned

6.13 Verification, commissioning, release, and change control (VAL) - 27 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB				
VAL-001-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB VAL-001-02	Use documented analysis, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; A/ T; High; Planned
VVC-SUB VAL-001-03	Use controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I; High; Planned
VVC-SUB VAL-002-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SUB VAL-002-02	Use objective testing, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/ A; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VAL-002-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-SUB				
VAL-003-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; Gate; Planned
VVC-SUB				
VAL-003-02	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; Moderate; Planned
VVC-SUB				
VAL-003-03	Use documented analysis, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; A/I; Moderate; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VAL-004-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SUB				
VAL-004-02	Use objective testing, witnessed demonstration to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/D; High; Planned
VVC-SUB				
VAL-004-03	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SUB				
VAL-005-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; Gate; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VAL-005-02	Use documented analysis, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; A/ T; High; Planned
VVC-SUB				
VAL-005-03	Use controlled inspection, representative-user human-factors evaluation to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/HFE; High; Planned
VVC-SUB				
VAL-006-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SUB				
VAL-006-02	Use controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VAL-006-03	Use documented analysis, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; A/ I; High; Planned
VVC-SUB				
VAL-007-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SUB				
VAL-007-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T; High; Planned
VVC-SUB				
VAL-007-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VAL-008-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SUB				
VAL-008-02	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SUB				
VAL-008-03	Use objective testing, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/I; High; Planned
VVC-SUB				
VAL-009-01	Use controlled inspection, documented analysis to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SUB				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VAL-009-02	Use objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T; High; Planned
VVC-SUB				
VAL-009-03	Use witnessed demonstration, controlled inspection to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Demonstrate this child within a representative controlled reference-case, anomaly, regression, and release-decision scenario and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; D/I; High; Planned

6.14 Accreditation-profile and clinical-quality governance (ACC) - 24 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SUB				
ACC-001-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
ACC-001-02	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SUB				
ACC-001-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SUB				
ACC-002-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SUB				
ACC-002-02	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
ACC-002-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SUB				
ACC-003-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SUB				
ACC-003-02	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SUB				
ACC-003-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
ACC-004-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SUB				
ACC-004-02	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SUB				
ACC-004-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SUB				
ACC-005-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
ACC-005-02	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SUB				
ACC-005-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SUB				
ACC-006-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SUB				
ACC-006-02	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
ACC-006-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SUB				
ACC-007-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SUB				
ACC-007-02	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SUB				
ACC-007-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SUB				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
ACC-008-01	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SUB				
ACC-008-02	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SUB				
ACC-008-03	Use controlled inspection, objective testing to execute the exact normative statement through every allocated component. Include positive, negative, boundary, stale, incompatible, failure, recovery, and audit cases applicable to source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Demonstrate this child within a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims and confirm that its output supports the parent outcome without unintended state, authority, or workflow effects.	The exact expected result is obtained for every mandatory case and every allocated component path is covered. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned

7 High-level interface requirement V&V checks

Each HLIR check covers the paired producer-consumer contract, integrated route, assigned interface families, and all three SIR children.

7.1 Governance, intended use, and scope (GOV) - 21 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -GOV -001-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-HLIR				
IF -GOV -001-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/D; Moderate; Planned
VVC-HLIR				
IF -GOV -001-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T/A; Moderate; Planned
VVC-HLIR				
IF -GOV -002-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-HLIR				
IF -GOV -002-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/D; Moderate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -GOV -002-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T/A; Moderate; Planned
VVC-HLIR				
IF -GOV -003-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-HLIR				
IF -GOV -003-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/D; Moderate; Planned
VVC-HLIR				
IF -GOV -003-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T/A; Moderate; Planned
VVC-HLIR				
IF -GOV -004-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-HLIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -GOV -004-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/D; Moderate; Planned
VVC-HLIR				
IF -GOV -004-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T/A; Moderate; Planned
VVC-HLIR				
IF -GOV -005-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-HLIR				
IF -GOV -005-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/D; Moderate; Planned
VVC-HLIR				
IF -GOV -005-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T/A; Moderate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -GOV -006-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-HLIR				
IF -GOV -006-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/D; Moderate; Planned
VVC-HLIR				
IF -GOV -006-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T/A; Moderate; Planned
VVC-HLIR				
IF -GOV -007-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-HLIR				
IF -GOV -007-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/D; Moderate; Planned
VVC-HLIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -GOV -007-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge scope, authority, separation of duties, effective configuration, change, and release, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific governed scope, change, approval, and release scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T/A; Moderate; Planned

7.2 Safety boundaries and human authority (SAF) - 30 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF -SAF -001-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-HLIR IF -SAF -001-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-HLIR IF -SAF -001-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T/A; High; Planned
VVC-HLIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -SAF -002-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-HLIR				
IF -SAF -002-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-HLIR				
IF -SAF -002-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T/A; High; Planned
VVC-HLIR				
IF -SAF -003-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-SAF-003-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-HLIR				
IF-SAF-003-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T/A; High; Planned
VVC-HLIR				
IF-SAF-004-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-HLIR				
IF-SAF-004-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-HLIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-SAF-004-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T/A; High; Planned
VVC-HLIR				
IF-SAF-005-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-HLIR				
IF-SAF-005-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-HLIR				
IF-SAF-005-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -SAF -006-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-HLIR				
IF -SAF -006-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-HLIR				
IF -SAF -006-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T/A; High; Planned
VVC-HLIR				
IF -SAF -007-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-SAF-007-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-HLIR				
IF-SAF-007-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T/A; High; Planned
VVC-HLIR				
IF-SAF-008-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-HLIR				
IF-SAF-008-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-SAF-008-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T/A; High; Planned
VVC-HLIR				
IF-SAF-009-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-HLIR				
IF-SAF-009-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-HLIR				
IF-SAF-009-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -SAF -010-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-HLIR				
IF -SAF -010-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-HLIR				
IF -SAF -010-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific safety-critical clinical workflow with wrong-context and prohibited-action challenges; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T/A; High; Planned

7.3 Natural-language and voice interaction (NLI) - 24 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR				
IF -NLI -001-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-HLIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -NLI-001-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-HLIR				
IF -NLI-001-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T/A; High; Planned
VVC-HLIR				
IF -NLI-002-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-HLIR				
IF -NLI-002-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-HLIR				
IF -NLI-002-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T/A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF-NLI-003-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-HLIR IF-NLI-003-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-HLIR IF-NLI-003-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T/A; High; Planned
VVC-HLIR IF-NLI-004-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-HLIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -NLI-004-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-HLIR				
IF -NLI-004-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T/A; High; Planned
VVC-HLIR				
IF -NLI-005-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-HLIR				
IF -NLI-005-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-HLIR				
IF -NLI-005-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T/A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF-NLI-006-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-HLIR IF-NLI-006-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-HLIR IF-NLI-006-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T/A; High; Planned
VVC-HLIR IF-NLI-007-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -NLI-007-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-HLIR				
IF -NLI-007-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T/A; High; Planned
VVC-HLIR				
IF -NLI-008-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-HLIR				
IF -NLI-008-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-HLIR				
IF -NLI-008-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific typed and push-to-talk workflow containing realistic ambiguity and critical tokens; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T/A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
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7.4 Evidence and literature governance (EVD) - 24 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
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VVC-HLIR

IF -EVD-001-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
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VVC-HLIR

IF -EVD-001-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/D; High; Planned
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VVC-HLIR

IF -EVD-001-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T/A; High; Planned
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VVC-HLIR

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -EVD -002-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-HLIR				
IF -EVD -002-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/D; High; Planned
VVC-HLIR				
IF -EVD -002-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T/A; High; Planned
VVC-HLIR				
IF -EVD -003-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -EVD -003-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/D; High; Planned
VVC-HLIR				
IF -EVD -003-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T/A; High; Planned
VVC-HLIR				
IF -EVD -004-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-HLIR				
IF -EVD -004-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/D; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -EVD -004 -03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ T/A; High; Planned
VVC-HLIR				
IF -EVD -005 -01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ A; High; Planned
VVC-HLIR				
IF -EVD -005 -02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/ D; High; Planned
VVC-HLIR				
IF -EVD -005 -03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ T/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -EVD -006 -01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-HLIR				
IF -EVD -006 -02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/D; High; Planned
VVC-HLIR				
IF -EVD -006 -03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T/A; High; Planned
VVC-HLIR				
IF -EVD -007 -01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -EVD -007-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/D; High; Planned
VVC-HLIR				
IF -EVD -007-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T/A; High; Planned
VVC-HLIR				
IF -EVD -008-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-HLIR				
IF -EVD -008-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/D; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-EVD-008-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T/A; High; Planned

7.5 Clinical context, imaging, registration, and contours (CLN) - 27 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR				
IF-CLN-001-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-HLIR				
IF-CLN-001-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-HLIR				
IF-CLN-001-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T/A; High; Planned

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF-CLN-002-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-HLIR IF-CLN-002-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-HLIR IF-CLN-002-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T/A; High; Planned
VVC-HLIR IF-CLN-003-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-CLN-003-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-HLIR				
IF-CLN-003-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T/A; High; Planned
VVC-HLIR				
IF-CLN-004-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-HLIR				
IF-CLN-004-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-CLN-004-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T/A; High; Planned
VVC-HLIR				
IF-CLN-005-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-HLIR				
IF-CLN-005-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-HLIR				
IF-CLN-005-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-CLN-006-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-HLIR				
IF-CLN-006-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-HLIR				
IF-CLN-006-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T/A; High; Planned
VVC-HLIR				
IF-CLN-007-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-CLN-007-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-HLIR				
IF-CLN-007-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T/A; High; Planned
VVC-HLIR				
IF-CLN-008-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-HLIR				
IF-CLN-008-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-HLIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-CLN-008-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T/A; High; Planned
VVC-HLIR				
IF-CLN-009-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-HLIR				
IF-CLN-009-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-HLIR				
IF-CLN-009-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative imaging, registration, contouring, and anatomy-review case; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T/A; High; Planned

7.6 Treatment-plan generation and calculation (PLN) - 36 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF -PLN-001-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-HLIR IF -PLN-001-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-HLIR IF -PLN-001-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T/A; High; Planned
VVC-HLIR IF -PLN-002-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -PLN -002 -02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-HLIR				
IF -PLN -002 -03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T/A; High; Planned
VVC-HLIR				
IF -PLN -003 -01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-HLIR				
IF -PLN -003 -02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -PLN -003 -03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T/A; High; Planned
VVC-HLIR				
IF -PLN -004 -01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-HLIR				
IF -PLN -004 -02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-HLIR				
IF -PLN -004 -03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-PLN-005-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-HLIR				
IF-PLN-005-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-HLIR				
IF-PLN-005-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T/A; High; Planned
VVC-HLIR				
IF-PLN-006-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-PLN-006-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-HLIR				
IF-PLN-006-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T/A; High; Planned
VVC-HLIR				
IF-PLN-007-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-HLIR				
IF-PLN-007-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -PLN-007-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T/A; High; Planned
VVC-HLIR				
IF -PLN-008-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-HLIR				
IF -PLN-008-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-HLIR				
IF -PLN-008-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T/A; High; Planned
VVC-HLIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-PLN-009-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-HLIR				
IF-PLN-009-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-HLIR				
IF-PLN-009-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T/A; High; Planned
VVC-HLIR				
IF-PLN-010-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-HLIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-PLN-010-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-HLIR				
IF-PLN-010-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T/A; High; Planned
VVC-HLIR				
IF-PLN-011-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-HLIR				
IF-PLN-011-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-HLIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-PLN-011-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T/A; High; Planned
VVC-HLIR				
IF-PLN-012-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-HLIR				
IF-PLN-012-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-HLIR				
IF-PLN-012-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative treatment-planning case spanning intent, candidates, dose, and robustness; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T/A; High; Planned

7.7 Plan review, selection, QA, and transfer (REV) - 27 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF-REV-001-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-HLIR IF-REV-001-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/D; Gate; Planned
VVC-HLIR IF-REV-001-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T/A; High; Planned
VVC-HLIR IF-REV-002-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-REV-002-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/D; Gate; Planned
VVC-HLIR				
IF-REV-002-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T/A; High; Planned
VVC-HLIR				
IF-REV-003-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-HLIR				
IF-REV-003-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/D; Gate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -REV -003-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T/A; High; Planned
VVC-HLIR				
IF -REV -004-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ A; High; Planned
VVC-HLIR				
IF -REV -004-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/ D; Gate; Planned
VVC-HLIR				
IF -REV -004-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-REV-005-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-HLIR				
IF-REV-005-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/D; Gate; Planned
VVC-HLIR				
IF-REV-005-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T/A; High; Planned
VVC-HLIR				
IF-REV-006-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-REV-006-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/D; Gate; Planned
VVC-HLIR				
IF-REV-006-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T/A; High; Planned
VVC-HLIR				
IF-REV-007-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-HLIR				
IF-REV-007-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/D; Gate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -REV -007 -03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T/A; High; Planned
VVC-HLIR				
IF -REV -008 -01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ A; High; Planned
VVC-HLIR				
IF -REV -008 -02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/ D; Gate; Planned
VVC-HLIR				
IF -REV -008 -03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-REV-009-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-HLIR				
IF-REV-009-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/D; Gate; Planned
VVC-HLIR				
IF-REV-009-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T/A; High; Planned

7.8 Data, interoperability, provenance, and audit (DAT) - 30 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR				
IF-DAT-001-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF-DAT-001-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-HLIR IF-DAT-001-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T/A; High; Planned
VVC-HLIR IF-DAT-002-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-HLIR IF-DAT-002-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-DAT-002-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T/A; High; Planned
VVC-HLIR				
IF-DAT-003-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-HLIR				
IF-DAT-003-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-HLIR				
IF-DAT-003-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-DAT-004-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-HLIR				
IF-DAT-004-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-HLIR				
IF-DAT-004-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T/A; High; Planned
VVC-HLIR				
IF-DAT-005-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-DAT-005-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-HLIR				
IF-DAT-005-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T/A; High; Planned
VVC-HLIR				
IF-DAT-006-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-HLIR				
IF-DAT-006-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-DAT-006-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T/A; High; Planned
VVC-HLIR				
IF-DAT-007-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-HLIR				
IF-DAT-007-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-HLIR				
IF-DAT-007-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-DAT-008-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-HLIR				
IF-DAT-008-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-HLIR				
IF-DAT-008-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T/A; High; Planned
VVC-HLIR				
IF-DAT-009-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-DAT-009-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-HLIR				
IF-DAT-009-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T/A; High; Planned
VVC-HLIR				
IF-DAT-010-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-HLIR				
IF-DAT-010-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-DAT-010-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific clinical-object lifecycle and round-trip data exchange workflow; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T/A; High; Planned

7.9 AI and inference-model lifecycle (AIM) - 24 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR				
IF-AIM-001-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-HLIR				
IF-AIM-001-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/D; High; Planned
VVC-HLIR				
IF-AIM-001-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T/A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF - AIM - 002 - 01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-HLIR IF - AIM - 002 - 02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/D; High; Planned
VVC-HLIR IF - AIM - 002 - 03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T/A; High; Planned
VVC-HLIR IF - AIM - 003 - 01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF - AIM - 003 - 02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/D; High; Planned
VVC-HLIR				
IF - AIM - 003 - 03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T/A; High; Planned
VVC-HLIR				
IF - AIM - 004 - 01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-HLIR				
IF - AIM - 004 - 02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/D; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF - AIM - 004 - 03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T/A; High; Planned
VVC-HLIR				
IF - AIM - 005 - 01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-HLIR				
IF - AIM - 005 - 02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/D; High; Planned
VVC-HLIR				
IF - AIM - 005 - 03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF - AIM - 006 - 01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-HLIR				
IF - AIM - 006 - 02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/D; High; Planned
VVC-HLIR				
IF - AIM - 006 - 03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T/A; High; Planned
VVC-HLIR				
IF - AIM - 007 - 01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF - AIM - 007 - 02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/D; High; Planned
VVC-HLIR				
IF - AIM - 007 - 03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T/A; High; Planned
VVC-HLIR				
IF - AIM - 008 - 01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-HLIR				
IF - AIM - 008 - 02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/D; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-AIM-008-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific model-supported clinical draft workflow including abstention, failure, and monitoring cases; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T/A; High; Planned

7.10 Human factors, usability, and competency (HFE) - 18 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR				
IF-HFE-001-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-HLIR				
IF-HFE-001-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/D; Moderate; Planned
VVC-HLIR				
IF-HFE-001-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T/A; Moderate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -HFE-002-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-HLIR				
IF -HFE-002-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/D; Moderate; Planned
VVC-HLIR				
IF -HFE-002-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T/A; Moderate; Planned
VVC-HLIR				
IF -HFE-003-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -HFE-003-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/D; Moderate; Planned
VVC-HLIR				
IF -HFE-003-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T/A; Moderate; Planned
VVC-HLIR				
IF -HFE-004-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-HLIR				
IF -HFE-004-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/D; Moderate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -HFE-004-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T/A; Moderate; Planned
VVC-HLIR				
IF -HFE-005-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-HLIR				
IF -HFE-005-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/D; Moderate; Planned
VVC-HLIR				
IF -HFE-005-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T/A; Moderate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -HFE-006-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-HLIR				
IF -HFE-006-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/D; Moderate; Planned
VVC-HLIR				
IF -HFE-006-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific representative critical-task usability scenario with interruptions and recovery; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T/A; Moderate; Planned

7.11 Privacy, cybersecurity, and secure development (SEC) - 24 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -SEC-001-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -SEC -001-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/D; High; Planned
VVC-HLIR				
IF -SEC -001-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T/A; High; Planned
VVC-HLIR				
IF -SEC -002-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-HLIR				
IF -SEC -002-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/D; High; Planned
VVC-HLIR				
IF -SEC -002-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T/A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF-SEC-003-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-HLIR IF-SEC-003-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/D; High; Planned
VVC-HLIR IF-SEC-003-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T/A; High; Planned
VVC-HLIR IF-SEC-004-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -SEC -004 -02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/D; High; Planned
VVC-HLIR				
IF -SEC -004 -03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T/A; High; Planned
VVC-HLIR				
IF -SEC -005 -01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-HLIR				
IF -SEC -005 -02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/D; High; Planned
VVC-HLIR				
IF -SEC -005 -03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T/A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF-SEC-006-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-HLIR IF-SEC-006-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/D; High; Planned
VVC-HLIR IF-SEC-006-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T/A; High; Planned
VVC-HLIR IF-SEC-007-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -SEC -007 -02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/D; High; Planned
VVC-HLIR				
IF -SEC -007 -03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T/A; High; Planned
VVC-HLIR				
IF -SEC -008 -01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-HLIR				
IF -SEC -008 -02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/D; High; Planned
VVC-HLIR				
IF -SEC -008 -03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific threat, privacy, access-control, and incident-response scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T/A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
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7.12 Clinical operations, monitoring, and resilience (OPS) - 21 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF-OPS-001-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-HLIR IF-OPS-001-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-HLIR IF-OPS-001-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T/A; Moderate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-OPS-002-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-HLIR				
IF-OPS-002-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-HLIR				
IF-OPS-002-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T/A; Moderate; Planned
VVC-HLIR				
IF-OPS-003-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-OPS-003-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-HLIR				
IF-OPS-003-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T/A; Moderate; Planned
VVC-HLIR				
IF-OPS-004-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-HLIR				
IF-OPS-004-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-OPS-004-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T/A; Moderate; Planned
VVC-HLIR				
IF-OPS-005-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-HLIR				
IF-OPS-005-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-HLIR				
IF-OPS-005-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T/A; Moderate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-OPS-006-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-HLIR				
IF-OPS-006-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-HLIR				
IF-OPS-006-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T/A; Moderate; Planned
VVC-HLIR				
IF-OPS-007-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-OPS-007-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-HLIR				
IF-OPS-007-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific production-like load, dependency failure, degraded-mode, and recovery scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T/A; Moderate; Planned

7.13 Verification, commissioning, release, and change control (VAL) - 27 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR				
IF-VAL-001-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-HLIR				
IF-VAL-001-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/D; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF-VAL-001-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T/A; High; Planned
VVC-HLIR IF-VAL-002-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-HLIR IF-VAL-002-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/D; High; Planned
VVC-HLIR IF-VAL-002-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-VAL-003-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-HLIR				
IF-VAL-003-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/D; High; Planned
VVC-HLIR				
IF-VAL-003-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T/A; High; Planned
VVC-HLIR				
IF-VAL-004-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -VAL-004-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/D; High; Planned
VVC-HLIR				
IF -VAL-004-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T/A; High; Planned
VVC-HLIR				
IF -VAL-005-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-HLIR				
IF -VAL-005-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/D; High; Planned
VVC-HLIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-VAL-005-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T/A; High; Planned
VVC-HLIR				
IF-VAL-006-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-HLIR				
IF-VAL-006-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/D; High; Planned
VVC-HLIR				
IF-VAL-006-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-VAL-007-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-HLIR				
IF-VAL-007-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/D; High; Planned
VVC-HLIR				
IF-VAL-007-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T/A; High; Planned
VVC-HLIR				
IF-VAL-008-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF -VAL-008-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/D; High; Planned
VVC-HLIR				
IF -VAL-008-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T/A; High; Planned
VVC-HLIR				
IF -VAL-009-01	Use controlled inspection, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-HLIR				
IF -VAL-009-02	Use objective testing, witnessed demonstration to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/D; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-VAL-009-03	Use controlled inspection, objective testing, documented analysis to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific controlled reference-case, anomaly, regression, and release-decision scenario; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; A; High; Planned

7.14 Accreditation-profile and clinical-quality governance (ACC) - 24 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF-ACC-001-01	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-HLIR IF-ACC-001-02	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-HLIR IF-ACC-001-03	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-HLIR IF - ACC -002-01	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-HLIR IF - ACC -002-02	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-HLIR IF - ACC -002-03	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-HLIR IF - ACC -003-01	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF - ACC - 003 - 02	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-HLIR				
IF - ACC - 003 - 03	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-HLIR				
IF - ACC - 004 - 01	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-HLIR				
IF - ACC - 004 - 02	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF - ACC - 004 - 03	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-HLIR				
IF - ACC - 005 - 01	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-HLIR				
IF - ACC - 005 - 02	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-HLIR				
IF - ACC - 005 - 03	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF - ACC - 006 - 01	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-HLIR				
IF - ACC - 006 - 02	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-HLIR				
IF - ACC - 006 - 03	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-HLIR				
IF - ACC - 007 - 01	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-HLIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF - ACC - 007 - 02	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-HLIR				
IF - ACC - 007 - 03	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-HLIR				
IF - ACC - 008 - 01	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-HLIR				
IF - ACC - 008 - 02	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-HLIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IF-ACC-008-03	Use controlled inspection, objective testing to inspect producer and consumer contracts and execute the integrated route across every assigned interface family. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation, partial results, and reconciliation.	Commission a representative site-, version-, machine-, and route-specific QMP-led quality and accreditation-evidence workflow without system-generated compliance claims; producer, consumer, clinical owner, security, operations, and T-VV confirm fitness.	All three SIR checks and every assigned interface-family profile pass paired producer-consumer and site-route acceptance. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned

8 Sub-interface requirement V&V checks

Each SIR check verifies its assigned realization package at every relevant boundary and validates its contribution within the parent route.

8.1 Governance, intended use, and scope (GOV) - 63 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-GOV-001-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; Moderate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-001-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; Moderate; Planned
VVC-SIR				
SIR-GOV-001-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/D; Moderate; Planned
VVC-SIR				
SIR-GOV-001-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-001-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-001-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				
SIR-GOV-001-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-002-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; Moderate; Planned
VVC-SIR				
SIR-GOV-002-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ A; Moderate; Planned
VVC-SIR				
SIR-GOV-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; Moderate; Planned
VVC-SIR				
SIR-GOV-002-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/ D; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-002-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				
SIR-GOV-002-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-002-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				
SIR-GOV-002-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-003-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-003-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; Moderate; Planned
VVC-SIR				
SIR-GOV-003-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-003-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-003-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/D; Moderate; Planned
VVC-SIR				
SIR-GOV-003-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				
SIR-GOV-003-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-003-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-003-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-004-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-004-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; Moderate; Planned
VVC-SIR				
SIR-GOV-004-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-004-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; Moderate; Planned
VVC-SIR				
SIR-GOV-004-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/D; Moderate; Planned
VVC-SIR				
SIR-GOV-004-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				
SIR-GOV-004-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-004-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				
SIR-GOV-004-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-005-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-005-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-005-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-005-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; Moderate; Planned
VVC-SIR				
SIR-GOV-005-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/D; Moderate; Planned
VVC-SIR				
SIR-GOV-005-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-005-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-005-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				
SIR-GOV-005-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-006-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; Moderate; Planned
VVC-SIR				
SIR-GOV-006-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ A; Moderate; Planned
VVC-SIR				
SIR-GOV-006-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; Moderate; Planned
VVC-SIR				
SIR-GOV-006-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/ D; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-006-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				
SIR-GOV-006-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-006-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				
SIR-GOV-006-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-007-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-007-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; Moderate; Planned
VVC-SIR				
SIR-GOV-007-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-007-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-007-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/D; Moderate; Planned
VVC-SIR				
SIR-GOV-007-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				
SIR-GOV-007-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned
VVC-SIR				
SIR-GOV-007-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; T/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-GOV-007-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge scope, authority, separation of duties, effective configuration, change, and release and unsafe exception paths.	Validate the package contribution within its parent route during a representative governed scope, change, approval, and release scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/A; Moderate; Planned

8.2 Safety boundaries and human authority (SAF) - 90 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-SAF-001-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-SIR				
SIR-SAF-001-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR SIR-SAF-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; Gate; Planned
VVC-SIR SIR-SAF-001-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/ D; Gate; Planned
VVC-SIR SIR-SAF-001-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/ A; Gate; Planned
VVC-SIR SIR-SAF-001-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-001-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; High; Planned
VVC-SIR				
SIR-SAF-001-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-002-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-002-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Gate; Planned
VVC-SIR				
SIR-SAF-002-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-SIR				
SIR-SAF-002-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-002-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-002-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; High; Planned
VVC-SIR				
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VVC-SIR				
SIR-SAF-003-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-003-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; High; Planned
VVC-SIR				
SIR-SAF-003-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ A; High; Planned
VVC-SIR				
SIR-SAF-003-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; Gate; Planned
VVC-SIR				
SIR-SAF-003-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/ D; Gate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-003-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; Gate; Planned
VVC-SIR				
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VVC-SIR				
SIR-SAF-003-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; High; Planned
VVC-SIR				
SIR-SAF-003-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-004-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-004-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-SIR				
SIR-SAF-004-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-004-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-004-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-SIR				
SIR-SAF-004-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; Gate; Planned
VVC-SIR				
SIR-SAF-004-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-004-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-004-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-005-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
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VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-005-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; Gate; Planned
VVC-SIR				
SIR-SAF-005-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/ D; Gate; Planned
VVC-SIR				
SIR-SAF-005-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/ A; Gate; Planned
VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-005-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; High; Planned
VVC-SIR				
SIR-SAF-005-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-006-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-006-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-006-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Gate; Planned
VVC-SIR				
SIR-SAF-006-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-SIR				
SIR-SAF-006-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-006-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-006-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; High; Planned
VVC-SIR				
SIR-SAF-006-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-007-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-007-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; High; Planned
VVC-SIR				
SIR-SAF-007-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ A; High; Planned
VVC-SIR				
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VVC-SIR				
SIR-SAF-007-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/ D; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-007-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; Gate; Planned
VVC-SIR				
SIR-SAF-007-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
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VVC-SIR				
SIR-SAF-007-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-008-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-008-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-SIR				
SIR-SAF-008-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-008-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-008-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-SIR				
SIR-SAF-008-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; Gate; Planned
VVC-SIR				
SIR-SAF-008-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-008-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-008-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-009-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-009-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-SIR				
SIR-SAF-009-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-009-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Gate; Planned
VVC-SIR				
SIR-SAF-009-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-SIR				
SIR-SAF-009-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; Gate; Planned
VVC-SIR				
SIR-SAF-009-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-009-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; High; Planned
VVC-SIR				
SIR-SAF-009-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-010-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-010-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-010-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/A; High; Planned
VVC-SIR				
SIR-SAF-010-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Gate; Planned
VVC-SIR				
SIR-SAF-010-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/D; Gate; Planned
VVC-SIR				
SIR-SAF-010-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/A; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SAF-010-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ A; High; Planned
VVC-SIR				
SIR-SAF-010-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; T/ A; High; Planned
VVC-SIR				
SIR-SAF-010-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative safety-critical clinical workflow with wrong-context and prohibited-action challenges, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ A; High; Planned

8.3 Natural-language and voice interaction (NLI) - 72 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-NLI-001-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/ A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-NLI-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/ T; High; Planned
VVC-SIR				
SIR-NLI-001-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/ A; High; Planned
VVC-SIR				
SIR-NLI-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/ T; High; Planned
VVC-SIR				
SIR-NLI-001-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/ D; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-001-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				
SIR-NLI-001-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-001-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				
SIR-NLI-001-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-002-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				
SIR-NLI-002-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-002-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-SIR				
SIR-NLI-002-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				
SIR-NLI-002-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-002-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-002-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-003-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-003-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				
SIR-NLI-003-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-003-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				
SIR-NLI-003-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-SIR				
SIR-NLI-003-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				
SIR-NLI-003-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-003-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				
SIR-NLI-003-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-004-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-004-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-004-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-004-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				
SIR-NLI-004-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-SIR				
SIR-NLI-004-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-004-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-004-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				
SIR-NLI-004-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-005-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-005-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				
SIR-NLI-005-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-005-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				
SIR-NLI-005-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-005-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				
SIR-NLI-005-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-005-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				
SIR-NLI-005-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-006-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				
SIR-NLI-006-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-006-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-006-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-SIR				
SIR-NLI-006-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				
SIR-NLI-006-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-006-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-006-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-007-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-007-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				
SIR-NLI-007-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-007-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				
SIR-NLI-007-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-SIR				
SIR-NLI-007-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				
SIR-NLI-007-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-007-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				
SIR-NLI-007-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-008-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-008-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-008-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-008-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-SIR				
SIR-NLI-008-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/D; High; Planned
VVC-SIR				
SIR-NLI-008-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-NLI-008-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned
VVC-SIR				
SIR-NLI-008-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; T/A; High; Planned
VVC-SIR				
SIR-NLI-008-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation and unsafe exception paths.	Validate the package contribution within its parent route during a representative typed and push-to-talk workflow containing realistic ambiguity and critical tokens, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/A; High; Planned

8.4 Evidence and literature governance (EVD) - 72 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-EVD-001-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-EVD-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SIR				
SIR-EVD-001-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SIR				
SIR-EVD-001-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/D; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-001-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				
SIR-EVD-001-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-001-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				
SIR-EVD-001-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-002-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SIR				
SIR-EVD-002-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-002-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/D; High; Planned
VVC-SIR				
SIR-EVD-002-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				
SIR-EVD-002-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-002-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-002-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-003-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-003-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SIR				
SIR-EVD-003-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-003-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ T; High; Planned
VVC-SIR				
SIR-EVD-003-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/ D; High; Planned
VVC-SIR				
SIR-EVD-003-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/ A; High; Planned
VVC-SIR				
SIR-EVD-003-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-003-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				
SIR-EVD-003-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-004-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-004-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-004-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-004-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SIR				
SIR-EVD-004-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/D; High; Planned
VVC-SIR				
SIR-EVD-004-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-004-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-004-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				
SIR-EVD-004-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-005-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-005-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ T; High; Planned
VVC-SIR				
SIR-EVD-005-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ A; High; Planned
VVC-SIR				
SIR-EVD-005-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ T; High; Planned
VVC-SIR				
SIR-EVD-005-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/ D; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-005-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				
SIR-EVD-005-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-005-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				
SIR-EVD-005-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-006-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SIR				
SIR-EVD-006-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-006-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-006-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/D; High; Planned
VVC-SIR				
SIR-EVD-006-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				
SIR-EVD-006-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-006-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-006-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-007-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-007-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-007-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ T; High; Planned
VVC-SIR				
SIR-EVD-007-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/ D; High; Planned
VVC-SIR				
SIR-EVD-007-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/ A; High; Planned
VVC-SIR				
SIR-EVD-007-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-007-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				
SIR-EVD-007-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-008-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-008-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-008-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-008-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-SIR				
SIR-EVD-008-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/D; High; Planned
VVC-SIR				
SIR-EVD-008-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-EVD-008-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned
VVC-SIR				
SIR-EVD-008-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; T/A; High; Planned
VVC-SIR				
SIR-EVD-008-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical evidence selection and computable-rule workflow with conflict, expiry, and missing-rule cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/A; High; Planned

8.5 Clinical context, imaging, registration, and contours (CLN) - 81 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-CLN-001-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-CLN-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; High; Planned
VVC-SIR				
SIR-CLN-001-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-SIR				
SIR-CLN-001-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-001-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; Gate; Planned
VVC-SIR				
SIR-CLN-001-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-001-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; High; Planned
VVC-SIR				
SIR-CLN-001-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-002-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; High; Planned
VVC-SIR				
SIR-CLN-002-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-002-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-SIR				
SIR-CLN-002-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; Gate; Planned
VVC-SIR				
SIR-CLN-002-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-002-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-002-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-003-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-003-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; High; Planned
VVC-SIR				
SIR-CLN-003-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-003-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-SIR				
SIR-CLN-003-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-SIR				
SIR-CLN-003-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; Gate; Planned
VVC-SIR				
SIR-CLN-003-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-003-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; High; Planned
VVC-SIR				
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VVC-SIR				
SIR-CLN-004-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
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VVC-SIR				
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VVC-SIR				
SIR-CLN-004-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-SIR				
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VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-004-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-004-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; High; Planned
VVC-SIR				
SIR-CLN-004-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-005-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-005-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; High; Planned
VVC-SIR				
SIR-CLN-005-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-005-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-SIR				
SIR-CLN-005-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-005-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; Gate; Planned
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VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-006-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; High; Planned
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VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-006-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
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VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-006-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
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VVC-SIR				
SIR-CLN-007-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/ D; Gate; Planned
VVC-SIR				
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VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-007-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; High; Planned
VVC-SIR				
SIR-CLN-007-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-008-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-008-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-008-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
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VVC-SIR				
SIR-CLN-008-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-SIR				
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VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-008-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
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VVC-SIR				
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VVC-SIR				
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VVC-SIR				

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SIR-CLN-009-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-009-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-SIR				
SIR-CLN-009-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/D; Gate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-CLN-009-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; Gate; Planned
VVC-SIR				
SIR-CLN-009-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/A; High; Planned
VVC-SIR				
SIR-CLN-009-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative imaging, registration, contouring, and anatomy-review case, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; T/A; High; Planned
VVC-SIR				
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8.6 Treatment-plan generation and calculation (PLN) - 108 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-PLN-001-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				
SIR-PLN-001-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-001-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-SIR				
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VVC-SIR				
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VVC-SIR				
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VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-001-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-002-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				
SIR-PLN-002-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-SIR				
SIR-PLN-002-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-SIR				
SIR-PLN-002-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; Gate; Planned
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VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-002-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; High; Planned
VVC-SIR				
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VVC-SIR				
SIR-PLN-003-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
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VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-003-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
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VVC-SIR				
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VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
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VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
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VVC-SIR				

Continued on next page

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SIR-PLN-004-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; High; Planned
VVC-SIR				
SIR-PLN-004-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-005-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-005-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				
SIR-PLN-005-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-005-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-005-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-SIR				
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VVC-SIR				
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VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-005-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-006-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
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VVC-SIR				

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SIR-PLN-006-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
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SIR-PLN-006-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-006-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; High; Planned
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VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-007-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
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VVC-SIR				
SIR-PLN-007-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-007-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
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VVC-SIR				
SIR-PLN-007-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-008-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-008-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				
SIR-PLN-008-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
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VVC-SIR				
SIR-PLN-008-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-008-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; Gate; Planned
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-009-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-SIR				
SIR-PLN-009-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; Gate; Planned
VVC-SIR				
SIR-PLN-009-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-009-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-009-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-010-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-010-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				
SIR-PLN-010-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-010-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-SIR				
SIR-PLN-010-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-SIR				
SIR-PLN-010-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; Gate; Planned
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-010-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; High; Planned
VVC-SIR				
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VVC-SIR				
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VVC-SIR				
SIR-PLN-011-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-011-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
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VVC-SIR				
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VVC-SIR				
SIR-PLN-011-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; Gate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-011-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-011-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; High; Planned
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-012-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				
SIR-PLN-012-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-012-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-SIR				
SIR-PLN-012-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/D; Gate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-PLN-012-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; Gate; Planned
VVC-SIR				
SIR-PLN-012-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/A; High; Planned
VVC-SIR				
SIR-PLN-012-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative treatment-planning case spanning intent, candidates, dose, and robustness, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; T/A; High; Planned
VVC-SIR				
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8.7 Plan review, selection, QA, and transfer (REV) - 81 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-REV-001-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-SIR				
SIR-REV-001-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; Gate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-001-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/D; Gate; Planned
VVC-SIR				
SIR-REV-001-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; Gate; Planned
VVC-SIR				
SIR-REV-001-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-001-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-001-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-002-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-SIR				
SIR-REV-002-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; Gate; Planned
VVC-SIR				
SIR-REV-002-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/ D; Gate; Planned
VVC-SIR				
SIR-REV-002-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/ A; Gate; Planned
VVC-SIR				
SIR-REV-002-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-002-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; High; Planned
VVC-SIR				
SIR-REV-002-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-003-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-003-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-003-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-003-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; Gate; Planned
VVC-SIR				
SIR-REV-003-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/D; Gate; Planned
VVC-SIR				
SIR-REV-003-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-003-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-003-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; High; Planned
VVC-SIR				
SIR-REV-003-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-004-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-004-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SIR				
SIR-REV-004-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ A; High; Planned
VVC-SIR				
SIR-REV-004-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; Gate; Planned
VVC-SIR				
SIR-REV-004-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/ D; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-004-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; Gate; Planned
VVC-SIR				
SIR-REV-004-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-004-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; High; Planned
VVC-SIR				
SIR-REV-004-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-005-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-005-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-SIR				
SIR-REV-005-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-005-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; Gate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-005-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/D; Gate; Planned
VVC-SIR				
SIR-REV-005-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; Gate; Planned
VVC-SIR				
SIR-REV-005-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-005-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-005-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-006-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-SIR				
SIR-REV-006-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-006-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; Gate; Planned
VVC-SIR				
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VVC-SIR				
SIR-REV-006-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/ A; Gate; Planned
VVC-SIR				
SIR-REV-006-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-006-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; High; Planned
VVC-SIR				
SIR-REV-006-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-007-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-007-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-007-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-007-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; Gate; Planned
VVC-SIR				
SIR-REV-007-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/D; Gate; Planned
VVC-SIR				
SIR-REV-007-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-007-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-007-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; High; Planned
VVC-SIR				
SIR-REV-007-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-008-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-008-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SIR				
SIR-REV-008-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ A; High; Planned
VVC-SIR				
SIR-REV-008-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; Gate; Planned
VVC-SIR				
SIR-REV-008-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/ D; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-008-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; Gate; Planned
VVC-SIR				
SIR-REV-008-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-008-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; High; Planned
VVC-SIR				
SIR-REV-008-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-009-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-009-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-SIR				
SIR-REV-009-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-009-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-009-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/D; Gate; Planned
VVC-SIR				
SIR-REV-009-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; Gate; Planned
VVC-SIR				
SIR-REV-009-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned
VVC-SIR				
SIR-REV-009-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-REV-009-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative multidisciplinary plan comparison, approval, QA, transfer, and reconciliation workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/A; High; Planned

8.8 Data, interoperability, provenance, and audit (DAT) - 90 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-DAT-001-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SIR				
SIR-DAT-001-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR SIR-DAT-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-SIR SIR-DAT-001-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-SIR SIR-DAT-001-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; Gate; Planned
VVC-SIR SIR-DAT-001-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-001-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; High; Planned
VVC-SIR				
SIR-DAT-001-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-002-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-002-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and approvals and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-SIR				
SIR-DAT-002-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-SIR				
SIR-DAT-002-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-002-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-002-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; High; Planned
VVC-SIR				
SIR-DAT-002-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-003-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-003-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SIR				
SIR-DAT-003-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-003-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-SIR				
SIR-DAT-003-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-003-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; Gate; Planned
VVC-SIR				
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VVC-SIR				
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VVC-SIR				
SIR-DAT-003-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-004-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-004-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SIR				
SIR-DAT-004-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-004-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-004-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-SIR				
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VVC-SIR				
SIR-DAT-004-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-004-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-005-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
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VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-005-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-SIR				
SIR-DAT-005-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-SIR				
SIR-DAT-005-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; Gate; Planned
VVC-SIR				
SIR-DAT-005-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-005-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; High; Planned
VVC-SIR				
SIR-DAT-005-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-006-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-006-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-006-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and approvals and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-SIR				
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VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-006-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-006-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and approvals and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; High; Planned
VVC-SIR				
SIR-DAT-006-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-007-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-007-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SIR				
SIR-DAT-007-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-007-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-SIR				
SIR-DAT-007-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-007-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; Gate; Planned
VVC-SIR				
SIR-DAT-007-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-007-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; High; Planned
VVC-SIR				
SIR-DAT-007-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-008-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-008-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SIR				
SIR-DAT-008-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-008-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-008-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-SIR				
SIR-DAT-008-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; Gate; Planned
VVC-SIR				
SIR-DAT-008-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-008-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-008-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-009-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-009-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SIR				
SIR-DAT-009-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-009-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-SIR				
SIR-DAT-009-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-SIR				
SIR-DAT-009-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; Gate; Planned
VVC-SIR				
SIR-DAT-009-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-009-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; High; Planned
VVC-SIR				
SIR-DAT-009-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-010-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-010-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-010-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-010-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and approvals and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-SIR				
SIR-DAT-010-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/D; Gate; Planned
VVC-SIR				
SIR-DAT-010-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; Gate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-DAT-010-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned
VVC-SIR				
SIR-DAT-010-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; T/A; High; Planned
VVC-SIR				
SIR-DAT-010-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative clinical-object lifecycle and round-trip data exchange workflow, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/A; High; Planned

8.9 AI and inference-model lifecycle (AIM) - 72 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-AIM-001-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-AIM-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T; High; Planned
VVC-SIR				
SIR-AIM-001-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SIR				
SIR-AIM-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T; High; Planned
VVC-SIR				
SIR-AIM-001-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/D; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-001-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/A; High; Planned
VVC-SIR				
SIR-AIM-001-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SIR				
SIR-AIM-001-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/A; High; Planned
VVC-SIR				
SIR-AIM-001-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-002-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ A; High; Planned
VVC-SIR				
SIR-AIM-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-SIR				
SIR-AIM-002-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ A; High; Planned
VVC-SIR				
SIR-AIM-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-002-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/D; High; Planned
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-002-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SIR				
SIR-AIM-003-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SIR				
SIR-AIM-003-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T; High; Planned
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-003-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-SIR				
SIR-AIM-003-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/ D; High; Planned
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-003-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/A; High; Planned
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VVC-SIR				
SIR-AIM-004-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-004-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ A; High; Planned
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VVC-SIR				
SIR-AIM-004-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/ D; High; Planned
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VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-004-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-005-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
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VVC-SIR				

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SIR-AIM-005-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/A; High; Planned
VVC-SIR				
SIR-AIM-005-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-006-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ A; High; Planned
VVC-SIR				
SIR-AIM-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-SIR				
SIR-AIM-006-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ A; High; Planned
VVC-SIR				
SIR-AIM-006-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-006-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/D; High; Planned
VVC-SIR				
SIR-AIM-006-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/A; High; Planned
VVC-SIR				
SIR-AIM-006-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SIR				
SIR-AIM-006-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-006-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SIR				
SIR-AIM-007-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SIR				
SIR-AIM-007-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T; High; Planned
VVC-SIR				
SIR-AIM-007-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-007-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-SIR				
SIR-AIM-007-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/ D; High; Planned
VVC-SIR				
SIR-AIM-007-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/ A; High; Planned
VVC-SIR				
SIR-AIM-007-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-007-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/A; High; Planned
VVC-SIR				
SIR-AIM-007-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SIR				
SIR-AIM-008-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SIR				
SIR-AIM-008-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-008-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ A; High; Planned
VVC-SIR				
SIR-AIM-008-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-SIR				
SIR-AIM-008-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/ D; High; Planned
VVC-SIR				
SIR-AIM-008-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/ A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-AIM-008-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned
VVC-SIR				
SIR-AIM-008-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; T/A; High; Planned
VVC-SIR				
SIR-AIM-008-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change and unsafe exception paths.	Validate the package contribution within its parent route during a representative model-supported clinical draft workflow including abstention, failure, and monitoring cases, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/A; High; Planned

8.10 Human factors, usability, and competency (HFE) - 54 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-HFE-001-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-HFE-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-SIR				
SIR-HFE-001-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-SIR				
SIR-HFE-001-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/D; Moderate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-HFE-001-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/A; Moderate; Planned
VVC-SIR				
SIR-HFE-001-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-001-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/A; Moderate; Planned
VVC-SIR				
SIR-HFE-001-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-HFE-002-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-SIR				
SIR-HFE-002-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-HFE-002-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/D; Moderate; Planned
VVC-SIR				
SIR-HFE-002-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/A; Moderate; Planned
VVC-SIR				
SIR-HFE-002-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-002-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-HFE-002-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-003-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-003-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-SIR				
SIR-HFE-003-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-HFE-003-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-SIR				
SIR-HFE-003-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/D; Moderate; Planned
VVC-SIR				
SIR-HFE-003-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/A; Moderate; Planned
VVC-SIR				
SIR-HFE-003-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-HFE-003-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/A; Moderate; Planned
VVC-SIR				
SIR-HFE-003-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-004-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-004-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-HFE-004-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-004-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-SIR				
SIR-HFE-004-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/D; Moderate; Planned
VVC-SIR				
SIR-HFE-004-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-HFE-004-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-004-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/A; Moderate; Planned
VVC-SIR				
SIR-HFE-004-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-005-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-HFE-005-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-SIR				
SIR-HFE-005-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-005-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-SIR				
SIR-HFE-005-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/D; Moderate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-HFE-005-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/A; Moderate; Planned
VVC-SIR				
SIR-HFE-005-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-005-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/A; Moderate; Planned
VVC-SIR				
SIR-HFE-005-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-HFE-006-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-SIR				
SIR-HFE-006-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-006-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-HFE-006-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/D; Moderate; Planned
VVC-SIR				
SIR-HFE-006-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/A; Moderate; Planned
VVC-SIR				
SIR-HFE-006-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned
VVC-SIR				
SIR-HFE-006-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; T/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-HFE-006-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery and unsafe exception paths.	Validate the package contribution within its parent route during a representative representative critical-task usability scenario with interruptions and recovery, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/A; Moderate; Planned

8.11 Privacy, cybersecurity, and secure development (SEC) - 72 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-SEC-001-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SIR				
SIR-SEC-001-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR SIR-SEC-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ T; High; Planned
VVC-SIR SIR-SEC-001-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/ D; High; Planned
VVC-SIR SIR-SEC-001-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/ A; High; Planned
VVC-SIR SIR-SEC-001-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-001-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				
SIR-SEC-001-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-002-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-002-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SIR				
SIR-SEC-002-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/D; High; Planned
VVC-SIR				
SIR-SEC-002-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-002-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-002-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				
SIR-SEC-002-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-003-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-003-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ T; High; Planned
VVC-SIR				
SIR-SEC-003-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ A; High; Planned
VVC-SIR				
SIR-SEC-003-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ T; High; Planned
VVC-SIR				
SIR-SEC-003-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/ D; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-003-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				
SIR-SEC-003-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-003-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				
SIR-SEC-003-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-004-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-004-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SIR				
SIR-SEC-004-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-004-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-004-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/D; High; Planned
VVC-SIR				
SIR-SEC-004-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				
SIR-SEC-004-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-004-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-004-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-005-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-005-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SIR				
SIR-SEC-005-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-005-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SIR				
SIR-SEC-005-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/D; High; Planned
VVC-SIR				
SIR-SEC-005-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				
SIR-SEC-005-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-005-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				
SIR-SEC-005-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-006-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-006-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-006-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SIR				
SIR-SEC-006-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/D; High; Planned
VVC-SIR				
SIR-SEC-006-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-006-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-006-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				
SIR-SEC-006-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-007-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-007-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ T; High; Planned
VVC-SIR				
SIR-SEC-007-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ A; High; Planned
VVC-SIR				
SIR-SEC-007-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/ T; High; Planned
VVC-SIR				
SIR-SEC-007-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/ D; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-007-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				
SIR-SEC-007-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-007-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				
SIR-SEC-007-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-008-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-008-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SIR				
SIR-SEC-008-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-008-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-008-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/D; High; Planned
VVC-SIR				
SIR-SEC-008-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				
SIR-SEC-008-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned
VVC-SIR				
SIR-SEC-008-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; T/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-SEC-008-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response and unsafe exception paths.	Validate the package contribution within its parent route during a representative threat, privacy, access-control, and incident-response scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/A; High; Planned

8.12 Clinical operations, monitoring, and resilience (OPS) - 63 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-OPS-001-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				
SIR-OPS-001-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-OPS-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				
SIR-OPS-001-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-SIR				
SIR-OPS-001-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; Moderate; Planned
VVC-SIR				
SIR-OPS-001-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-001-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; Moderate; Planned
VVC-SIR				
SIR-OPS-001-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-002-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-002-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				
SIR-OPS-002-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-SIR				
SIR-OPS-002-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-002-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-002-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; Moderate; Planned
VVC-SIR				
SIR-OPS-002-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-003-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-003-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				
SIR-OPS-003-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-003-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				
SIR-OPS-003-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-003-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; Moderate; Planned
VVC-SIR				
SIR-OPS-003-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
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VVC-SIR				
SIR-OPS-003-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-004-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-004-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				
SIR-OPS-004-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-004-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-004-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-SIR				
SIR-OPS-004-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; Moderate; Planned
VVC-SIR				
SIR-OPS-004-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-004-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-005-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-005-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				
SIR-OPS-005-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-005-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				
SIR-OPS-005-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-SIR				
SIR-OPS-005-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; Moderate; Planned
VVC-SIR				
SIR-OPS-005-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-005-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; Moderate; Planned
VVC-SIR				
SIR-OPS-005-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-006-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-006-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-006-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				
SIR-OPS-006-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-006-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-006-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; Moderate; Planned
VVC-SIR				
SIR-OPS-006-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-007-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-007-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				
SIR-OPS-007-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-007-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-SIR				
SIR-OPS-007-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/D; Moderate; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-OPS-007-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; Moderate; Planned
VVC-SIR				
SIR-OPS-007-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned
VVC-SIR				
SIR-OPS-007-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; T/A; Moderate; Planned
VVC-SIR				
SIR-OPS-007-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation and unsafe exception paths.	Validate the package contribution within its parent route during a representative production-like load, dependency failure, degraded-mode, and recovery scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/A; Moderate; Planned

8.13 Verification, commissioning, release, and change control (VAL) - 81 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-VAL-001-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SIR				
SIR-VAL-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-SIR				
SIR-VAL-001-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SIR				
SIR-VAL-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-001-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/D; High; Planned
VVC-SIR				
SIR-VAL-001-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/A; High; Planned
VVC-SIR				
SIR-VAL-001-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SIR				
SIR-VAL-001-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-001-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
SIR-VAL-002-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
SIR-VAL-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				
SIR-VAL-002-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				
SIR-VAL-002-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/ D; High; Planned
VVC-SIR				
SIR-VAL-002-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/ A; High; Planned
VVC-SIR				
SIR-VAL-002-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-002-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/A; High; Planned
VVC-SIR				
SIR-VAL-002-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SIR				
SIR-VAL-003-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SIR				
SIR-VAL-003-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-003-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
SIR-VAL-003-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				
SIR-VAL-003-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/ D; High; Planned
VVC-SIR				
SIR-VAL-003-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/ A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-003-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
SIR-VAL-003-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/ A; High; Planned
VVC-SIR				
SIR-VAL-003-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
SIR-VAL-004-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-004-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				
SIR-VAL-004-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
SIR-VAL-004-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				
SIR-VAL-004-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/ D; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-004-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/A; High; Planned
VVC-SIR				
SIR-VAL-004-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SIR				
SIR-VAL-004-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/A; High; Planned
VVC-SIR				
SIR-VAL-004-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-005-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
SIR-VAL-005-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-005-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/D; High; Planned
VVC-SIR				
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VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-005-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
SIR-VAL-006-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
SIR-VAL-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				
SIR-VAL-006-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-006-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				
SIR-VAL-006-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/ D; High; Planned
VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-006-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/A; High; Planned
VVC-SIR				
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VVC-SIR				
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VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-007-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
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VVC-SIR				
SIR-VAL-007-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/ D; High; Planned
VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-007-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
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VVC-SIR				
SIR-VAL-007-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
SIR-VAL-008-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-008-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				
SIR-VAL-008-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
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VVC-SIR				
SIR-VAL-008-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/ D; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-008-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/A; High; Planned
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VVC-SIR				
SIR-VAL-008-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/A; High; Planned
VVC-SIR				
SIR-VAL-008-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-009-01-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
SIR-VAL-009-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				
SIR-VAL-009-01-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ A; High; Planned
VVC-SIR				
SIR-VAL-009-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-009-02-02	Use objective testing, witnessed demonstration to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/D; High; Planned
VVC-SIR				
SIR-VAL-009-02-03	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/A; High; Planned
VVC-SIR				
SIR-VAL-009-03-01	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned
VVC-SIR				
SIR-VAL-009-03-02	Use objective testing, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; T/A; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-VAL-009-03-03	Use controlled inspection, documented analysis to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence and unsafe exception paths.	Validate the package contribution within its parent route during a representative controlled reference-case, anomaly, regression, and release-decision scenario, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/A; High; Planned

8.14 Accreditation-profile and clinical-quality governance (ACC) - 72 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-ACC-001-01-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-001-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-001-01-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-SIR				
SIR-ACC-001-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-001-02-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-001-02-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-001-03-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-001-03-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-001-03-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-002-01-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-002-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-002-01-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-002-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-002-02-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-002-02-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-002-03-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-002-03-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-002-03-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-003-01-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-003-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-003-01-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-003-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-003-02-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-003-02-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-003-03-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-003-03-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-003-03-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-004-01-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-004-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-004-01-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-004-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-004-02-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-004-02-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-004-03-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-004-03-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-004-03-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-005-01-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SIR				
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VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-005-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-005-03-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/ T; High; Planned
VVC-SIR				
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VVC-SIR				
SIR-ACC-006-01-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				
SIR-ACC-006-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-006-01-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-SIR				
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VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-006-03-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
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VVC-SIR				
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VVC-SIR				
SIR-ACC-007-01-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-007-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
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VVC-SIR				
SIR-ACC-007-02-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-007-02-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-007-03-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-007-03-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				
SIR-ACC-007-03-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-008-01-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				
SIR-ACC-008-01-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				
SIR-ACC-008-01-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				
SIR-ACC-008-02-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-008-02-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				
SIR-ACC-008-02-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
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SIR-ACC-008-03-01	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				
SIR-ACC-008-03-02	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned
VVC-SIR				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
SIR-ACC-008-03-03	Use controlled inspection, objective testing to verify the assigned realization package at producer, boundary, consumer, audit, and reconciliation points. Challenge source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation and unsafe exception paths.	Validate the package contribution within its parent route during a representative QMP-led quality and accreditation-evidence workflow without system-generated compliance claims, including expected failure, user response, support, and recovery.	The realization package is complete; producer, boundary, consumer, audit, failure, and recovery results match the approved contract. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned

9 Core component V&V checks

One direct check is assigned to each of the 45 core component responsibilities.

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-COMP				
C-GOV-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-COMP				
C-QMS-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-COMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
C-REQ-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-COMP				
C-ARCH-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-COMP				
C-IAM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; Gate; Planned
VVC-COMP				
C-SAFE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Gate; Planned
VVC-COMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
C-STATE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-COMP				
C-UX-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation; critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; High; Planned
VVC-COMP				
C-HFE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation; critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; High; Planned
VVC-COMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
C-VOICE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation; critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-COMP				
C-INTENT-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation; critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-COMP				
C-WF-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-COMP				
C-EVID-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-COMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
C-RULE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-COMP				
C-CASE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-COMP				
C-REG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-COMP				
C-SEG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-COMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
C-PLAN-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-COMP				
C-DOSE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-COMP				
C-REV-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; Gate; Planned
VVC-COMP				
C-QA-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-COMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
C-DICOM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation; producer-consumer compatibility and route behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; Gate; Planned
VVC-COMP C-DICOM-02	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation; producer-consumer compatibility and route behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; Gate; Planned
VVC-COMP C-TPS-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation; producer-consumer compatibility and route behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; High; Planned
VVC-COMP C-FHIR-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; High; Planned
VVC-COMP				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
C-TERM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-COMP				
C-OBJ-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-COMP				
C-PROV-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-COMP				
C-AUD-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-COMP				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
C-CFG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; High; Planned
VVC-COMP				
C-MGW-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-COMP				
C-MREG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-COMP				
C-MVAL-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-COMP				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
C-SEC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-COMP				
C-MON-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-COMP				
C-JOB-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-COMP				
C-RES-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-COMP				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
C-INC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-COMP				
C-VV-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-COMP				
C-TEST-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-COMP				
C-SBX-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-COMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
C-ACC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-COMP				
C-DOC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-COMP				
C-ACC-02	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-COMP				
C-PBT-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned

10 Interface component V&V checks

One direct check is assigned to each of the 61 interface component responsibilities.

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-ICOMP IC-BND-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-ICOMP IC-ICD-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-ICOMP IC-CAP-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; High; Planned
VVC-ICOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-OWN-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; Gate; Planned
VVC-ICOMP				
IC-CFG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-ICOMP				
IC-RLS-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-ICOMP				
IC-CHG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-ICOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-SCH-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation; producer-consumer compatibility and route behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-ICOMP				
IC-SEM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation; producer-consumer compatibility and route behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-ICOMP				
IC-OBJ-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-ICOMP				
IC-EXM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; Moderate; Planned
VVC-ICOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-VER-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-ICOMP				
IC-GTW-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-ICOMP				
IC-CTX-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-ICOMP				
IC-REF-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-ICOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-POL-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; Gate; Planned
VVC-ICOMP				
IC-SAF-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Gate; Planned
VVC-ICOMP				
IC-INT-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation; producer-consumer compatibility and route behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-ICOMP				
IC-QRT-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-ICOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-ORC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-ICOMP				
IC-TXN-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Gate; Planned
VVC-ICOMP				
IC-IDM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-ICOMP				
IC-ACK-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-ICOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC -CMP -01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-ICOMP				
IC -TMD -01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-ICOMP				
IC -AUD -01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-ICOMP				
IC -PRV -01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-ICOMP				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-MON-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-ICOMP				
IC-REC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; High; Planned
VVC-ICOMP				
IC-ALT-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-ICOMP				
IC-BDR-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-ICOMP				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-TIM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-ICOMP				
IC-DCM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; Gate; Planned
VVC-ICOMP				
IC-DWP-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; Gate; Planned
VVC-ICOMP				
IC-RTP-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; Gate; Planned
VVC-ICOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-DVA-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; Gate; Planned
VVC-ICOMP				
IC-TPS-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation; producer-consumer compatibility and route behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; Gate; Planned
VVC-ICOMP				
IC-VAP-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; High; Planned
VVC-ICOMP				
IC-IHE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation; producer-consumer compatibility and route behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; High; Planned
VVC-ICOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-UIB-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/ T; High; Planned
VVC-ICOMP				
IC-NLG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/ T; High; Planned
VVC-ICOMP				
IC-EVG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/ T; High; Planned
VVC-ICOMP				
IC-IMG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/ T; Gate; Planned
VVC-ICOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-PLN-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-ICOMP				
IC-REV-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; Gate; Planned
VVC-ICOMP				
IC-MDG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/T; High; Planned
VVC-ICOMP				
IC-DAT-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Moderate; Planned
VVC-ICOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-IAM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; Gate; Planned
VVC-ICOMP				
IC-SEC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-ICOMP				
IC-PTM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-ICOMP				
IC-TRC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-ICOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-TST-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-ICOMP				
IC-ITL-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-ICOMP				
IC-COM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-ICOMP				
IC-HFE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation; critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/ T; High; Planned
VVC-ICOMP				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-AIE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-ICOMP				
IC-RPT-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-ICOMP				
IC-ACP-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-ICOMP				
IC-EVL-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/ T; High; Planned
VVC-ICOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
IC-CTG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-ICOMP				
IC-PBA-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; High; Planned

11 F/NF/O category component V&V checks

One direct check is assigned to each of the 78 category-scoped component responsibilities.

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-CCOMP				
FC-REQ-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-CCOMP				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
FC-ARCH-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-CCOMP				
FC-CFG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-CCOMP				
FC-PROV-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-CCOMP				
FC-VV-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-CCOMP				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
FC-IAM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; Gate; Planned
VVC-CCOMP				
FC-SAFE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Gate; Planned
VVC-CCOMP				
FC-STATE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-CCOMP				
FC-UX-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation; critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; High; Planned
VVC-CCOMP				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
FC-VOICE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation; critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-CCOMP				
FC-INTENT-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation; critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-NLI; I/T; High; Planned
VVC-CCOMP				
FC-WF-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-CCOMP				
FC-EVID-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
FC-RULE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving source authority, applicability, version, effective date, conflict, exception, citation, and no-applicable-rule behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-EVD; I/T; High; Planned
VVC-CCOMP				
FC-CASE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-CCOMP				
FC-REG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-CCOMP				
FC-SEG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-IMG; I/T; Gate; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
FC-PLAN-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-CCOMP				
FC-DOSE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving prescription, intent, objectives, constraints, machine and technique, optimization, dose, uncertainty, and robustness.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-PLAN; I/T; Gate; Planned
VVC-CCOMP				
FC-REV-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; Gate; Planned
VVC-CCOMP				
FC-QA-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; Gate; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
FC-DICOM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation; producer-consumer compatibility and route behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; Gate; Planned
VVC-CCOMP				
FC-TPS-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation; producer-consumer compatibility and route behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-INT; I/T; High; Planned
VVC-CCOMP				
FC-TERM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; Gate; Planned
VVC-CCOMP				
FC-OBJ-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
FC-MGW-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-CCOMP				
FC-MREG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-CCOMP				
FC-MVAL-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving model identity, intended-use scope, inputs, outputs, uncertainty, abstention, subgroup performance, drift, and change.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-AI; I/ T; High; Planned
VVC-CCOMP				
FC-JOB-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/ T; Moderate; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
FC-AUD-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-CCOMP NFC-REQ-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-CCOMP NFC-ARCH-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-CCOMP NFC-CFG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
NFC-MQN-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/ T; High; Planned
VVC-CCOMP				
NFC-VV-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/ T; High; Planned
VVC-CCOMP				
NFC-SAFE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; High; Planned
VVC-CCOMP				
NFC-STATE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient and object context, prohibited authority, deterministic hard stops, approval state, and fail-closed behavior.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/ T; High; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
NFC-IAM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-CCOMP				
NFC-CTX-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving patient identity, study and frame references, geometry, registration, contours, uncertainty, edit lineage, and review state.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; Moderate; Planned
VVC-CCOMP				
NFC-HFE-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation; critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-CCOMP				
NFC-UX-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving ambiguity, negation, numbers, units, laterality, transcript correction, typed intent, read-back, and cancellation; critical tasks, persistent context, warning comprehension, use error, workload, accessibility, training, and recovery.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; Moderate; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
NFC-ALERT-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-UX; I/T; High; Planned
VVC-CCOMP				
NFC-SEC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-CCOMP				
NFC-PRIV-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving identity, authorization, privacy, integrity, secrets, egress, malicious input, supply chain, detection, and response.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-CCOMP				
NFC-CRY-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
NFC-EGRESS-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-CCOMP				
NFC-INPUT-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-CCOMP				
NFC-ART-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-CCOMP				
NFC-SDL-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
NFC-AUD-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-CCOMP				
NFC-DATA-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-CCOMP				
NFC-INC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-CCOMP				
OC-REQ-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
OC - GOV - 01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-CCOMP				
OC - QMS - 01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-CCOMP				
OC - CFG - 01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SYS; I/T; High; Planned
VVC-CCOMP				
OC - AUD - 01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving canonical identity, version, units, coordinates, integrity, lineage, audit, retention, DICOM semantics, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-DATA; I/T; High; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
OC - VV - 01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; Moderate; Planned
VVC-CCOMP				
OC - SCOPE - 01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-CCOMP				
OC - MODE - 01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-CCOMP				
OC - ACCESS - 01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-SEC; I/T; High; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
OC-REL-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-CCOMP				
OC-CHG-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-CCOMP				
OC-MON-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-CCOMP				
OC-ALERT-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
OC-INC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-CCOMP				
OC-CAPA-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; Moderate; Planned
VVC-CCOMP				
OC-BKP-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; High; Planned
VVC-CCOMP				
OC-DOWN-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-CCOMP				

Continued on next page

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
OC-REC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving atomic jobs, availability, health, alerts, partial failure, degraded operation, backup, restore, recovery, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-OPS; I/T; Moderate; Planned
VVC-CCOMP				
OC-COMM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-CCOMP				
OC-TEST-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving bidirectional trace, reference and challenge cases, expected results, anomaly closure, independence, regression, and release evidence.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-CCOMP				
OC-ANOM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
OC-REV-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving candidate identity, comparison basis, selection, approval, independent QA, transfer eligibility, acknowledgment, and reconciliation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-CCOMP				
FC-ACC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-CCOMP				
FC-DOC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-REV; I/T; High; Planned
VVC-CCOMP				
NFC-CLM-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving contract completeness, configuration, inputs, outputs, safe failure, monitoring, recovery, and traceability.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-GOV; I/T; High; Planned
VVC-CCOMP				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
OC-ACC-01	Use controlled inspection, objective testing to verify the component responsibility, executable contract, safe failure state, monitoring, rollback or recovery, and all allocated requirement paths. Fault-inject conditions involving scope, authority, separation of duties, effective configuration, change, and release; source-profile applicability, human accreditation authority, QMP ownership, evidence integrity, deficiencies, CAPA, and claim limitation.	Integrate the commissioned component in representative clinical and operational workflows; authorized users and operators confirm workflow fit, understandable state, human authority, safe failure, supportability, and recovery.	Every allocated requirement path, contract, unit, integration, fault, monitoring, and recovery check assigned to the component passes. Configuration and data hashes, expected and actual results, logs, audit, anomaly dispositions, and approvals are retained; PASS is prohibited with an unresolved release blocker.	T-VV; I/T; High; Planned

12 Machine-QA realization V&V checks

These 56 checks supplement the 2,088-entity controlled baseline and trace MQA-PKG-01 into the same independent V&V and clinical-governance process. They do not create automated machine-readiness or return-to-service authority.

12.1 MQA realization requirements - 12 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-MQA-R MQA-001	Close all three MQA child checks, then use controlled inspection, objective testing to verify the complete realization outcome. Challenge source allowlisting, read-only acquisition, integrity, atomic import, and reconciliation without allowing analytics or AI to adjudicate readiness.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All three direct MQA child checks and the parent end-to-end outcome pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-DATA/T-OPS; I/T; Gate; Planned
VVC-MQA-R MQA-002	Close all three MQA child checks, then use controlled inspection, objective testing to verify the complete realization outcome. Challenge canonical identity, version, units, machine and configuration scope, and lifecycle state without allowing analytics or AI to adjudicate readiness.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All three direct MQA child checks and the parent end-to-end outcome pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-DATA; I/T; Gate; Planned
VVC-MQA-R				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
MQA-003	Close all three MQA child checks, then use controlled inspection, objective testing to verify the complete realization outcome. Challenge approved protocol and tolerance applicability, effective time, uncertainty, and no-rule behavior without allowing analytics or AI to adjudicate readiness.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All three direct MQA child checks and the parent end-to-end outcome pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-EVD; I/T; Gate; Planned
VVC-MQA-R MQA-004	Close all three MQA child checks, then use controlled inspection, objective testing, documented analysis to verify the complete realization outcome. Challenge bidirectional provenance from source through calculation, review, approval, and readiness effect without allowing analytics or AI to adjudicate readiness.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All three direct MQA child checks and the parent end-to-end outcome pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-DATA/T-VV; I/T/A; Gate; Planned
VVC-MQA-R MQA-005	Close all three MQA child checks, then use objective testing, documented analysis to verify the complete realization outcome. Challenge missing, incomplete, unreadable, unsupported, conflicting, and protocol-era evidence without allowing analytics or AI to adjudicate readiness.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All three direct MQA child checks and the parent end-to-end outcome pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-OPS; T/A; Gate; Planned
VVC-MQA-R MQA-006	Close all three MQA child checks, then use objective testing, documented analysis, representative-user human-factors evaluation to verify the complete realization outcome. Challenge separation of source results from analytics, simulation, models, retrieval, and readiness adjudication without allowing analytics or AI to adjudicate readiness.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All three direct MQA child checks and the parent end-to-end outcome pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-SYS/T-VV; T/A/HFE; Gate; Planned
VVC-MQA-R				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
MQ A-007	Close all three MQA child checks, then use objective testing, controlled inspection to verify the complete realization outcome. Challenge QMP authority, approval identity, separation of duties, expiry, and return to service without allowing analytics or AI to adjudicate readiness.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All three direct MQA child checks and the parent end-to-end outcome pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-GOV; T/I; Gate; Planned
VVC-MQA-R MQ A-008	Close all three MQA child checks, then use controlled inspection, witnessed demonstration, objective testing to verify the complete realization outcome. Challenge deviation, containment, investigation, CAPA, retest, effectiveness review, and closure without allowing analytics or AI to adjudicate readiness.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All three direct MQA child checks and the parent end-to-end outcome pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-GOV; I/D/T; Gate; Planned
VVC-MQA-R MQ A-009	Close all three MQA child checks, then use objective testing, witnessed demonstration to verify the complete realization outcome. Challenge signed minimal readiness state, scope, invalidation, hold, expiry, and fail-closed consumption without allowing analytics or AI to adjudicate readiness.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All three direct MQA child checks and the parent end-to-end outcome pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-SYS; T/D; Gate; Planned
VVC-MQA-R MQ A-010	Close all three MQA child checks, then use objective testing, representative-user human-factors evaluation to verify the complete realization outcome. Challenge typed machine-QA query and draft workflow intents, read-back, confirmation, and prohibited authority without allowing analytics or AI to adjudicate readiness.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All three direct MQA child checks and the parent end-to-end outcome pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-NLI/T-UX; T/HFE; Gate; Planned
VVC-MQA-R				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
MQA-011	Close all three MQA child checks, then use controlled inspection, objective testing to verify the complete realization outcome. Challenge commissioned DICOM and TPS association, UID and frame identity, round trip, and reconciliation without allowing analytics or AI to adjudicate readiness.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All three direct MQA child checks and the parent end-to-end outcome pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-INT/QMP; I/T; Gate; Planned
VVC-MQA-R MQA-012	Close all three MQA child checks, then use controlled inspection, objective testing, witnessed demonstration to verify the complete realization outcome. Challenge threat analysis, privacy, acceptance, commissioning, regression, recovery, monitoring, and release without allowing analytics or AI to adjudicate readiness.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All three direct MQA child checks and the parent end-to-end outcome pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-VV/QMP; I/T/D; Gate; Planned

12.2 MQA sub-requirements - 36 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-MQA-S MQA-001-01	Use controlled inspection, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to source allowlisting, read-only acquisition, integrity, atomic import, and reconciliation.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-DATA/T-OPS; I/A; Gate; Planned
VVC-MQA-S MQA-001-02	Use objective testing, witnessed demonstration to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to source allowlisting, read-only acquisition, integrity, atomic import, and reconciliation.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-DATA/T-OPS; T/D; Gate; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-MQA-S MQA-001-03	Use objective testing, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to source allowlisting, read-only acquisition, integrity, atomic import, and reconciliation.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-DATA/T-OPS; T/A; Gate; Planned
VVC-MQA-S MQA-002-01	Use controlled inspection, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to canonical identity, version, units, machine and configuration scope, and lifecycle state.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-DATA; I/A; Gate; Planned
VVC-MQA-S MQA-002-02	Use objective testing, witnessed demonstration to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to canonical identity, version, units, machine and configuration scope, and lifecycle state.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-DATA; T/D; Gate; Planned
VVC-MQA-S MQA-002-03	Use objective testing, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to canonical identity, version, units, machine and configuration scope, and lifecycle state.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-DATA; T/A; Gate; Planned
VVC-MQA-S				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
MQA-003-01	Use controlled inspection, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to approved protocol and tolerance applicability, effective time, uncertainty, and no-rule behavior.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-EVD; I/A; Gate; Planned
VVC-MQA-S				
MQA-003-02	Use objective testing, witnessed demonstration to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to approved protocol and tolerance applicability, effective time, uncertainty, and no-rule behavior.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-EVD; T/D; Gate; Planned
VVC-MQA-S				
MQA-003-03	Use objective testing, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to approved protocol and tolerance applicability, effective time, uncertainty, and no-rule behavior.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-EVD; T/A; Gate; Planned
VVC-MQA-S				
MQA-004-01	Use controlled inspection, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to bidirectional provenance from source through calculation, review, approval, and readiness effect.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-DATA/T-VV; I/A; Gate; Planned
VVC-MQA-S				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
MQA-004-02	Use objective testing, witnessed demonstration to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to bidirectional provenance from source through calculation, review, approval, and readiness effect.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-DATA/T-VV; T/D; Gate; Planned
VVC-MQA-S				
MQA-004-03	Use objective testing, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to bidirectional provenance from source through calculation, review, approval, and readiness effect.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-DATA/T-VV; T/A; Gate; Planned
VVC-MQA-S				
MQA-005-01	Use controlled inspection, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to missing, incomplete, unreadable, unsupported, conflicting, and protocol-era evidence.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-OPS; I/A; Gate; Planned
VVC-MQA-S				
MQA-005-02	Use objective testing, witnessed demonstration to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to missing, incomplete, unreadable, unsupported, conflicting, and protocol-era evidence.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-OPS; T/D; Gate; Planned
VVC-MQA-S				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
MQA-005-03	Use objective testing, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to missing, incomplete, unreadable, unsupported, conflicting, and protocol-era evidence.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-OPS; T/A; Gate; Planned
VVC-MQA-S				
MQA-006-01	Use controlled inspection, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to separation of source results from analytics, simulation, models, retrieval, and readiness adjudication.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-SYS/T-VV; I/A; Gate; Planned
VVC-MQA-S				
MQA-006-02	Use objective testing, witnessed demonstration to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to separation of source results from analytics, simulation, models, retrieval, and readiness adjudication.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-SYS/T-VV; T/D; Gate; Planned
VVC-MQA-S				
MQA-006-03	Use objective testing, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to separation of source results from analytics, simulation, models, retrieval, and readiness adjudication.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-SYS/T-VV; T/A; Gate; Planned
VVC-MQA-S				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
MQA-007-01	Use controlled inspection, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to QMP authority, approval identity, separation of duties, expiry, and return to service.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-GOV; I/A; Gate; Planned
VVC-MQA-S				
MQA-007-02	Use objective testing, witnessed demonstration to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to QMP authority, approval identity, separation of duties, expiry, and return to service.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-GOV; T/D; Gate; Planned
VVC-MQA-S				
MQA-007-03	Use objective testing, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to QMP authority, approval identity, separation of duties, expiry, and return to service.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-GOV; T/A; Gate; Planned
VVC-MQA-S				
MQA-008-01	Use controlled inspection, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to deviation, containment, investigation, CAPA, retest, effectiveness review, and closure.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-GOV; I/A; Gate; Planned
VVC-MQA-S				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
MQA-008-02	Use objective testing, witnessed demonstration to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to deviation, containment, investigation, CAPA, retest, effectiveness review, and closure.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-GOV; T/D; Gate; Planned
VVC-MQA-S				
MQA-008-03	Use objective testing, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to deviation, containment, investigation, CAPA, retest, effectiveness review, and closure.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-GOV; T/A; Gate; Planned
VVC-MQA-S				
MQA-009-01	Use controlled inspection, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to signed minimal readiness state, scope, invalidation, hold, expiry, and fail-closed consumption.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-SYS; I/A; Gate; Planned
VVC-MQA-S				
MQA-009-02	Use objective testing, witnessed demonstration to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to signed minimal readiness state, scope, invalidation, hold, expiry, and fail-closed consumption.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-SYS; T/D; Gate; Planned
VVC-MQA-S				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
MQA-009-03	Use objective testing, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to signed minimal readiness state, scope, invalidation, hold, expiry, and fail-closed consumption.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	QMP/T-SYS; T/A; Gate; Planned
VVC-MQA-S				
MQA-010-01	Use controlled inspection, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to typed machine-QA query and draft workflow intents, read-back, confirmation, and prohibited authority.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-NLI/T-UX; I/A; Gate; Planned
VVC-MQA-S				
MQA-010-02	Use objective testing, witnessed demonstration to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to typed machine-QA query and draft workflow intents, read-back, confirmation, and prohibited authority.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-NLI/T-UX; T/D; Gate; Planned
VVC-MQA-S				
MQA-010-03	Use objective testing, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to typed machine-QA query and draft workflow intents, read-back, confirmation, and prohibited authority.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-NLI/T-UX; T/A; Gate; Planned
VVC-MQA-S				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
MQA-011-01	Use controlled inspection, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to commissioned DICOM and TPS association, UID and frame identity, round trip, and reconciliation.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-INT/QMP; I/A; Gate; Planned
VVC-MQA-S				
MQA-011-02	Use objective testing, witnessed demonstration to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to commissioned DICOM and TPS association, UID and frame identity, round trip, and reconciliation.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-INT/QMP; T/D; Gate; Planned
VVC-MQA-S				
MQA-011-03	Use objective testing, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to commissioned DICOM and TPS association, UID and frame identity, round trip, and reconciliation.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-INT/QMP; T/A; Gate; Planned
VVC-MQA-S				
MQA-012-01	Use controlled inspection, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to threat analysis, privacy, acceptance, commissioning, regression, recovery, monitoring, and release.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-VV/QMP; I/A; Gate; Planned
VVC-MQA-S				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
MQA-012-02	Use objective testing, witnessed demonstration to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to threat analysis, privacy, acceptance, commissioning, regression, recovery, monitoring, and release.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-VV/QMP; T/D; Gate; Planned
VVC-MQA-S				
MQA-012-03	Use objective testing, documented analysis to execute the exact MQA child obligation through its allocated components. Include source, boundary, missing, stale, wrong-context, failure, recovery, provenance, and audit cases applicable to threat analysis, privacy, acceptance, commissioning, regression, recovery, monitoring, and release.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	The direct expected result and every allocated component path pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-VV/QMP; T/A; Gate; Planned

12.3 MQA package subassemblies - 8 checks

Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-MQA-C				
MQA-A01	Use controlled inspection, objective testing, witnessed demonstration to verify the subassembly responsibility, contracts, allocations, safe state, monitoring, failure response, recovery, and trace coverage for machine-QA evidence, governance, readiness, language interaction, external association, and safe lifecycle behavior.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All allocated MQA requirement paths, contracts, unit, integration, fault, monitoring, recovery, and QMP workflow checks pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-VV/QMP; I/T/D; Gate; Planned
VVC-MQA-C				
MQA-A02	Use controlled inspection, objective testing, witnessed demonstration to verify the subassembly responsibility, contracts, allocations, safe state, monitoring, failure response, recovery, and trace coverage for machine-QA evidence, governance, readiness, language interaction, external association, and safe lifecycle behavior.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All allocated MQA requirement paths, contracts, unit, integration, fault, monitoring, recovery, and QMP workflow checks pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-VV/QMP; I/T/D; Gate; Planned

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
VVC-MQA-C MQA-A03	Use controlled inspection, objective testing, witnessed demonstration to verify the subassembly responsibility, contracts, allocations, safe state, monitoring, failure response, recovery, and trace coverage for machine-QA evidence, governance, readiness, language interaction, external association, and safe lifecycle behavior.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All allocated MQA requirement paths, contracts, unit, integration, fault, monitoring, recovery, and QMP workflow checks pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-VV/QMP; I/T/D; Gate; Planned
VVC-MQA-C MQA-A04	Use controlled inspection, objective testing, witnessed demonstration to verify the subassembly responsibility, contracts, allocations, safe state, monitoring, failure response, recovery, and trace coverage for machine-QA evidence, governance, readiness, language interaction, external association, and safe lifecycle behavior.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All allocated MQA requirement paths, contracts, unit, integration, fault, monitoring, recovery, and QMP workflow checks pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-VV/QMP; I/T/D; Gate; Planned
VVC-MQA-C MQA-A05	Use controlled inspection, objective testing, witnessed demonstration to verify the subassembly responsibility, contracts, allocations, safe state, monitoring, failure response, recovery, and trace coverage for machine-QA evidence, governance, readiness, language interaction, external association, and safe lifecycle behavior.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All allocated MQA requirement paths, contracts, unit, integration, fault, monitoring, recovery, and QMP workflow checks pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-VV/QMP; I/T/D; Gate; Planned
VVC-MQA-C MQA-A06	Use controlled inspection, objective testing, witnessed demonstration to verify the subassembly responsibility, contracts, allocations, safe state, monitoring, failure response, recovery, and trace coverage for machine-QA evidence, governance, readiness, language interaction, external association, and safe lifecycle behavior.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All allocated MQA requirement paths, contracts, unit, integration, fault, monitoring, recovery, and QMP workflow checks pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-VV/QMP; I/T/D; Gate; Planned
VVC-MQA-C				

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Check / entity	Verification check	Validation check	Acceptance evidence	Owner / method / gate
MQA-A07	Use controlled inspection, objective testing, witnessed demonstration to verify the subassembly responsibility, contracts, allocations, safe state, monitoring, failure response, recovery, and trace coverage for machine-QA evidence, governance, readiness, language interaction, external association, and safe lifecycle behavior.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All allocated MQA requirement paths, contracts, unit, integration, fault, monitoring, recovery, and QMP workflow checks pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-VV/QMP; I/T/D; Gate; Planned
VVC-MQA-C				
MQA-A08	Use controlled inspection, objective testing, witnessed demonstration to verify the subassembly responsibility, contracts, allocations, safe state, monitoring, failure response, recovery, and trace coverage for machine-QA evidence, governance, readiness, language interaction, external association, and safe lifecycle behavior.	A QMP-led representative Daily, Monthly, Annual, commissioning, maintenance, discrepancy, and return-to-service workflow confirms fitness, comprehensibility, human authority, safe failure, and integration with the NL-TPS and commissioned external routes.	All allocated MQA requirement paths, contracts, unit, integration, fault, monitoring, recovery, and QMP workflow checks pass. Source hashes, configuration, expected and actual results, audit, anomalies, QMP review, T-VV review, and release decision are retained; no unresolved readiness or release blocker remains.	T-VV/QMP; I/T/D; Gate; Planned

13 Execution sequence and release gates

1. Freeze the applicable requirements, risks, architecture, interfaces, component allocations, evidence, site, machines, models, data, software, and configuration manifest.
2. Approve check procedures, expected results, tolerance sources, datasets, environments, independence assignments, clinical validation protocols, and anomaly rules before execution.
3. Execute component and SIR checks first; preserve raw evidence and failed attempts. Resolve or formally block affected higher-level checks.
4. Execute sub-requirement and HLIR integration checks, including negative, boundary, concurrency, interoperability, failure, recovery, security, and human-factors cases.
5. Execute HLR and MQA parent validation using representative team-of-teams workflows in the intended environment; do not infer parent PASS solely from child counts.
6. Complete coverage analysis, anomaly and CAPA closure, residual-risk review, regression, site acceptance, QMP and clinical approvals, and independent T-VV release recommendation.
7. Clinical governance issues or withholds the scoped release. Production monitoring and post-release evidence feed change impact and regression.

14 Structural completeness audit

Audit item	Expected	Generated	Exceptions	Status
Controlled risk-baseline entity checks	2,088	2,088	0	Pass
MQA-PKG-01 supplemental checks	56	56	0	Pass
Unique check-to-entity pairs	2,144	2,144	0 duplicates	Pass
Categorized F/NF/O alias duplication	0	0	Aliases retained in source trace	Pass
Executed-result claims	0	0	All checks remain Planned	Pass

15 Definition of done

This V&V specification is realized only when each of the 2,144 checks has an approved executable procedure or controlled inspection/analysis record, complete trace, locked environment and evidence, expected and actual results, anomaly disposition, independent review, and applicable clinical/QMP validation. A release package shall prove complete bidirectional trace from ConOps through requirements, interfaces, components, risks, checks, results, anomalies, monitoring, and the exact scoped release decision. Until then, the matrix is a plan and no row is PASS.